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Korean Sentence Complexity Reduction for Machine Translation

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Abstract

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Text simplification used as a preprocessing task for the improved functionality of natural language processing systems has a long history of research based on European languages, yet, there is no research that has utilized Korean as the object of study. However, there is great demand for comprehensible Korean to English machine translations, yet due to the disparate nature of these two languages, machine translation often fails to achieve fluent results.

In order to improve the translation quality of Korean text as the source language, the first-ever rule-based Korean complexity reduction system was designed, constructed, and implemented in this study. This system was achieved by a unique technique termed "phrase-grouping and generalization of nuance structures," in Korean as a disambiguation tool. This technique has potential

applications in all languages and additional natural language processing tasks. On top of this, in order to set a foundation for which complexity reduction operations and combinations generate fluent Korean and improved machine translation output, a unique factorial approach to simplification generation was also implemented.

In order to assess the output of the system proposed in the current research, the parallel evaluation of simplified Korean text by Korean native speakers and the evaluation of translations by English native speakers was conducted. The translation systems used in this study were Google Translate and Moses, both statistical machine translation systems, and Naver Translate, a neural machine translation system. This is the first research to conduct experiments on the interaction of text simplification and neural networks. Additionally, no known research has analyzed output from three machine translation systems simultaneously.

Generally, the proposed system generated relatively fluent Korean, though due to the factorial nature by which simplifications were generated, sentence quality usually began to deteriorate after more than one simplification operation. On the other hand, the proposed system as a preprocessing task for machine translation consistently improved translation quality for all three systems utilized in this study by up to two performed simplifications.

In the case of the statistical machine translation systems used in this study, more than two simplifications deteriorated not only Korean sentence quality, but also translation quality. However, in the case of Naver Translate, the neural machine translation system used in this study, even three simplifications resulted in translation improvement according to the evaluators. This study, then, emphasizes the need for more research conducted on text simplification as the field of natural language processing transitions to neural network-based approaches and applications.

Keywords : text simplification, machine translation, Korean, natural language processing, machine learning, neural networks
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1. Introduction

Text simplification (TS) is a discipline that deals with the conversion of complicated text into a simpler alternative which bears the same meaning as the original text. Traditionally, TS is and was performed by humans, typically educators, who rewrite complicated text using simpler forms for the benefit of the learning impaired, those with literacy issues, or foreign language learners. However, in addition to being performed by humans, TS also has a long history of automation as a Natural Language Processing (NLP) task.

Within TS there are two main types of simplification: semantic or lexical simplification and syntactic simplification. Semantic simplification focuses on the conversion from difficult, infrequent words or phrases to a simpler alternative. For example, consider (1) below:

(1) a. President Barack Obama was given the gift of life in Kenya.

b. President Barack Obama was born in Kenya.

(1a) and (1b) ultimately convey the same meaning, however, (1b)'s *born* is a far more frequent, simple, and easily understood alternative to (1a)'s *given the gift of life*, and is therefore much easier to process for someone who struggles with esoteric vocabulary or phrasing. Syntactic simplification, on the other hand, focuses on the conversion of complex syntactic structures into simpler alternative(s). Please consider (2) below:

(2) a. President Obama is from Kenya and Donald Trump is from Mars.

b. President Obama is from Kenya. Donald Trump is from Mars.

(2a) is an example of two clauses being coordinated within the same sentence, resulting in a single complicated syntactic structure, while (2b) conveys the same meaning using two simpler structures.

TS, whether as a human or an automated NLP task, also has many applications within the NLP field itself. It is perhaps intuitive that TS could be helpful for humans, but it is not immediately intuitive that it would assist in an NLP environment. However, when one considers the sheer number of rules required to appropriately process normal language in a rule-based approach or that statistical machine learning is based primarily on frequency of occurrence, and it is a fact of language that shorter structures are simply more common than longer structures, it becomes a reasonable assumption the TS could help improve the functionality of NLP systems.

One such NLP task is machine translation (MT). MT, put simply, is the automated translation of one language to another, and MT has a long history of benefitting from TS. Indeed, as A. Siddharthan notes in his survey summarizing the field of TS, the combination of TS and MT is perhaps the most prolific application of automated TS (2014). Traditional MT is and was performed primarily via a rule-based (RMT) approach, before transitioning to the most prolific approach in modern MT: statistical machine translation (SMT). The benefits of TS on both versions of MT have been demonstrated throughout the years and will be discussed in detail later in Chapter 2, but it is noteworthy that regardless of the MT approach, TS can benefit both types of systems.

Cutting-edge MT systems, however, are based in the use of the deep learning technique known was neural networks. While the marriage of TS and neural network motivated MT (NMT) is still an extremely new undertaking with little-to-no known literature covering the topic, it is a goal of this study to address TS's future in MT as the field transitions to NMT.

Regardless of the MT approach, be it out-dated RMT or cutting-edge NMT, highly divergent language pairs pose problems that any MT system must overcome. Language pairs that share a common origin, structure, or evolution, for example, English and German, are capable of being translated relatively well by most MT systems. However, as Koehn points out in his book, even the most advanced statistical applications in SMT fall lamentably short of the goal when the language pairs are divergent in nature, for example, English and Korean (2009). Despite the underwhelming performance by most MT systems on disparate language pairs, it is in fact on these types of language pairs where there is the most demand for reliable MT systems.

Given that Korean to English MT is an intimidating undertaking for even the most modern and robust MT system, it stands to reason that a TS system could provide a valuable means by which to improve output. One shortcoming of TS, though, is its relative lack of exposure in Asian NLP. Indeed, at the time of writing, there has been very little known research conducted on the automation of TS on an Asian language, that is, one article on Japanese and one on Korean, however, both were created for the assistance of humans. The inclusion of Korean-based automated TS would then be a valuable contribution to nearly any NLP undertaking involving the language, and the techniques used in such a system could be extended to other related languages, such as Japanese or Mongolian.

1.1 Korean Text Simplification

As stated above, there is little known research on the simplification of Korean text as an automated NLP task. As such, there exists no foundation on which to base any research on Korean TS, nor a jumping-off point from which to expand. It would indeed be quite an undertaking to build a simplification system for an entire language from the ground up. That being the case, this research will instead focus its interests on the most basic version of syntactic simplification, which is the creation of independent sentences from clauses embedded, subordinated, or coordinated within a sentence. If we are able to demonstrate a reasonable and effective way of rephrasing these clauses into their own independent sentences, and demonstrate a method which can additionally improve MT output, it can then be left to future research for the expansion and adaption of a Korean TS system.

1.1.1 Korean Sentence Complexity Reduction

Sentence complexity, as defined by R Chandrasekar et al. and later redefined by A. Siddharthan, is based on the number of clauses a sentence bears (1996, 2002). The reason for this definition comes from the fact that all elements in a sentence are typically related in some way to the verb. For example, verbs can introduce arguments, bear case, gender, and number features, and supply their own content as well. As number of clauses in a sentence increases, it then becomes exceedingly more difficult of for an NLP system to process the information related to and introduced by these verbs. In terms of a rule-based approach, the longer and more complicated the sentence, the more rules needed to process the sentence. From a statistically motivated approach, as sentence length increases, so too does the number of calculations performed. MT, in particular, is done on a sentence by sentence basis, and as such, sentences with lower complexity are handled much more effectively by MT systems.

The most basic task then for any syntactic simplification system is the reduction of individual sentence complexity in a way that also maintains the original intended meaning of the sentence. As the vast majority of the TS literature is based in the simplification of European languages, we then must explore a wholly unique means by which to reduce Korean sentence complexity. On top of that, our system must not only output simplified Korean text which bears the same meaning as the original text, but must also contribute in a meaningful way to the improvement of MT system output. The specifics of such a system will be discussed in detail in Chapter 4.

1.2 Research Objectives

The first goal of this research is to expand on the field of text simplification with the inclusion of a syntactically motivated Korean simplification system. This is accomplished via the following steps taken in the implementation of our system:

(1) Complex Korean sentence splitting and simplification

(2) The creation of a factorial system for generating simplified sentences

(3) The grouping and generalization of nuance structures

It is our hope that the creation of such a system will set the foundation for future

TS research involving the Korean language.

The second goal is to demonstrate that Korean TS can have a positive influence on modern MT output. This goal will be accomplished by using two SMT systems, namely Moses and Google Translate, and one NMT system, Naver Translate. By demonstrating that Korean TS can assist in NLP tasks we hope to expose the field of TS to Asian NLP, which might then gain traction and lead to the simplification of other Asian languages. Additionally, we hope to demonstrate that, while neural networks are capable of great feats, TS can be a useful stepping stone between the current available results and desired results.

And finally we simply wish to note that while TS as a pre-processing task for NLP systems is very useful, TS also has place in the field of education. Korea has never been of greater interest to the world at large, and as interest in the country continues to grow, so too will the number of those wishing to learn the language. Unfortunately, Korean is not the most accessible language, and many find it difficult to learn, especially as sentences become longer and more complex. It is possible that TS, perhaps even automated TS, could be used a tool for helping learners of Korean transition from one stage in their language learning to another. Additionally native speakers of Korean who suffer from learning disabilities or struggle with literacy issues may also benefit from Korean text simplification.

1.3 Research Outline

The outline of this research is as follows: Chapter 2 provides a literature review of text simplification, machine translation, and TS and a preprocessing step for MT. Chapter 3 introduces and discusses the corpus used in this research.

Chapter 4 provides a detailed account of the simplification system we developed. Chapter 5 discusses the pilot study conducted using Moses and Google Translate. Chapter 6 contains the full-scale experiment using all three translation systems. Chapter 7 lays out the limitations of the current research, research we leave for the future, and concludes the paper.

2. Literature Review

In this chapter the relevant literature is reviewed concerning the history and development of text simplification, machine translation, and text simplification as a preprocessing step for machine translation. Additionally, in the machine translation section, the machine translation systems used in the current research will be introduced and briefly discussed.

2.1 Text Simplification

Text simplification (TS) can perhaps be best understood as the conversion of complex or normal written text to alternative(s) that use simpler or more common phrasing than the original text. It is of course imperative that during the conversion of the text, the original meaning of the text is preserved. TS has perhaps existed as a concept since the beginning of written language, however, its origin as a practiced and researched discipline can be traced back to work of Clark H. and Clark E. (1968). In their study, the Clarks documented and measured the semantic distinctions between words made by native speakers as the structure and content of sentences increase in complexity. This initial psychology research sparked a whole field of study, giving insight into the perception and understanding of written language of varying difficulty for both those with and without learning impairments.

It has been demonstrated throughout the decades in dozens of articles that the manual application of TS can have a profound benefit for those with literacy issues, such as dyslexia, allowing them to read and understand abstract concepts they would have otherwise been denied access to (Canning et al., 2000). Additionally, while the majority of the manual TS literature has used English as the object of study, the practice has spread to other European languages, such as Portuguese, and shown promising results for the learning impaired who speak other languages (Candido et al., 2009).

The automation of TS as a Natural Language Processing (NLP) task was first proposed by Chandrasekar et al., with the suggestion of specific rules to follow the next year (1996, 1997). Interestingly, Chandrasekar et al's initial proposal was not to create a TS system that would benefit the learning impaired, as had been the tradition in the field, but rather a simplifier that would make for speedier parsing of documents by NLP systems. Put simply, their proposal was to create a corpus with original and hand-simplified sentences aligned together, and then use a lightweight dependency analyzer to parse and learn rules from the aligned sentences. This early research did not make much progress as the author's goal was ultimately faster parsing, but the use of a parser to accomplish such a goal negated their efforts.

The next brick laid in the foundation of automated TS was done primarily in Dras' doctoral work. In his PhD dissertation, Mark Dras attempted to address wider groups of paraphrase options, employing synchronous TAG formalism and integer programming to create text conversions from constraints proposed on the system externally, such as length (1999). His contribution was motivated by English alone, but did indeed provide interesting solutions for pure syntactic rewrites. Please see below:

1. Light verb constructions:

   (a) Steven made an attempt to stop playing Hearts.

   (b) Steven attempted to stop playing Hearts.

2. Clausal Complements

   (a) His willingness to leave made Gillian upset.

   (b) He was willing to leave. This made Gillian upset.

3. Genitives

   (a) The arrival of the train

   (b) The train’s arrival

4. Cleft constructions

   (a) It was his best suit that John wore to the ball.

   (b) John wore his best suit to the ball.

While many TS researchers found Dras' approach too reliant on English grammar and too syntactically restrictive to be employed practically at the time, his use of constraint satisfaction through integer programming has enjoyed a bit of a renaissance in recent TS research (Angrosh et al., 2014). It is noteworthy, however, that Dras' approach supported both human and NLP-motivated, automated TS.

While the state of automated TS continued to advance for the next several years, it was A. Siddharthan that developed the basic architecture and text cohesion that set the standard and continues to enjoy prominence in modern TS (2002). The main goal behind Siddharthan's approach to TS was to create an implementable TS system that was not reliant on parsers, however, due to the flexibility and intuitive nature of Siddharthan's design, even with the use of a parser, this design was simple enough to implement. Not only that, but TS researchers have noted that when automated TS is brought into a language for the first time, most researchers base the foundation of their work on Siddharthan's structure (Specia et al., 2012). Though the majority of Siddharthan's early work was focused on syntactic simplification, lexical or semantic simplification was included in his architecture as well, shown below in Figure 2.1:

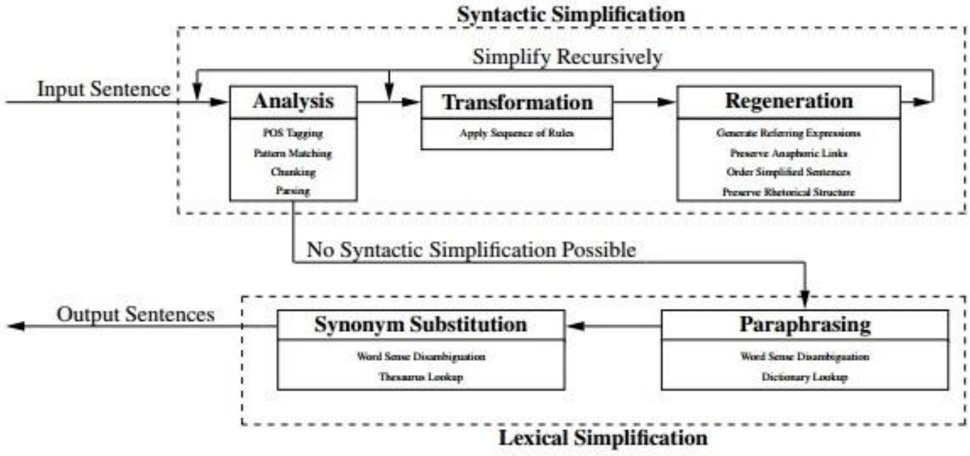


Figure 2.1: Siddharthan's Text Simplification Architecture

As can be seen in Figure 2.1, Siddharthan's approach was recursive in nature, performing all possible syntactic simplifications one at a time before transitioning to a lexical or semantic simplification stage. The reason for this being that a syntactic simplification may rule out a semantic simplification after being performed.

In the analysis stage of the architecture, the theoretical TS system performs or utilizes word level information, such part of speech tags, grammar structure pattern matching, chunking, and parsing. Additionally, it is in this stage where the system determines which, if any, simplification to perform. Based on the analysis from the first stage, the transformation stage begins simplifying the sentence in question, whether that be splitting the sentence into multiple sentences or converting a passive sentence into an active. It is in the next and final syntactic simplification stage where the simplified sentence(s) are given a cohesive logic, whether through the generation of sentence connecters, referring expressions, the generation or preservation of anaphors, or a sentence reordering. The system then returns the simplified sentence(s) to the initial stage where the process starts over again, should a simplification still be possible. If no simplification is possible, the simplified sentence(s) are sent to the lexical simplification stage.

The current research borrows heavily from the architecture suggested by Siddharthan for our purposed Korean syntactic TS system, however, due to the fact that Korean and English are disparate languages, our architecture is heavily altered and will be introduced in Chapter 4.

2.2 Machine Translation

Machine translation (MT) is perhaps the oldest NLP task, and can best be understood as the automatic conversion of text in one language to text in another. However, despite its long history, it is quite often the case that the output from an MT system is less than desirable. This is true for language pairs that are similar in structure, for example English and Spanish, but even more so for disparate language pairs, such as English and Korean. What follows will be a brief description of the most prominent MT system types and their relevance to the current research.

2.2.1 Rule-Based Machine Translation

Rule-Based Machine Translation (RMT) is the oldest form of MT, and is perhaps the most intuitive form of MT. Essentially, what a RMT system does is use linguistic information from a source language to convert it into a target language, making use of linguistic information from the target language as well (Hutchins, 1992). Essentially, what this means is the literal automated swapping of words or phrases between languages through the creation of rules. The approach is based in attempting to do computationally what the human mind does when translating a language. The benefits of an RMT system are quite extensive, as it really is the only known MT system that could potentially achieve 100 percent perfect translations. Unfortunately, such a system has never been achieved as the expense, not only in terms of computation but in sheer man-hours, is not only impractical; it is beyond imagination. As such, pure RMT systems have all but been abandoned in modern MT. Their only relevance to the current research is in the use of Hybrid MT systems, which make use of concepts from multiple MT system approaches. This will be discussed in detail later in section 2.2.4 as the current research could be viewed as a hybrid MT system.

2.2.2 Statistical Machine Translation

Statistical Machine Translation (SMT) is the most prolific form of modern MT. Essentially SMT assumes a statistical model for a given sentence in a source language. By thinking of translations as the combination of two models, a conditional translation model ($p(S|T)$) and a language model ($p(T)$), we are able assign and calculate probabilities, typically determined by N-grams (Koehn, 2009). The purpose of $p(S|T)$ calculation is to maintain meaning between the source and target language, while $p(T)$ is concerned with fluency of output in the target language (Na, 2015). Basically, these probabilities are calculated, a system is trained on a parallel bilingual corpus, and then a sentence in the source language is run through a decoder. Whatever translation in the target language yields the highest probability is the best translation, and this translation is what the SMT system in question outputs. In this way we can think about SMT as a Hidden Markov Model, where the possible translations are the hidden nodes and the best translation is the nodes whose combination yields the highest probability. Figure 2.2 below provides an adequate visualization of this process, taken from Li et al. (2012):

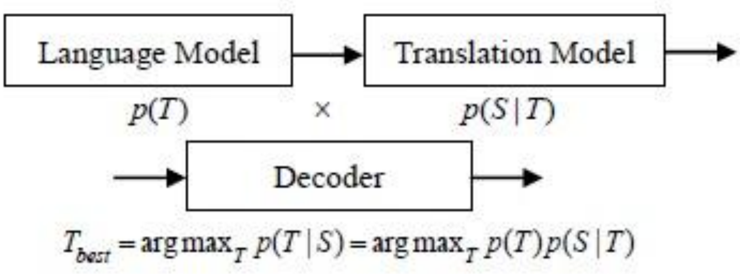


Figure 2.2: SMT visualization taken from Li et al.

SMT's prevalence in modern MT is in no short part due to the open-source toolkit for SMT, Moses (Koehn et al., 2007). Moses essentially functions as the decoder in Figure 2.2 above, providing a free, easily usable, trainable, and customizable environment where one might conduct SMT experiments. Though there are several other systems of comparable specifications with Moses, none has enjoyed the prominence Moses has, becoming the standard SMT system in the literature and typically acting as the baseline to determine improvement in SMT focused articles and studies.

Moses may be the most prominent SMT system involved in academic studies, but no SMT system enjoys more prominence and accessibility than Google Translate (2016). At the time of writing, Google Translate is a fully-trained, fully-functioning, freely accessible SMT system capable of translating between 103 language pairs, though more are in development (Ahmed, 2016).

Moses and Google Translate are based on the same SMT technology, however, Google Translate has no customizability options, nor can it be trained and changed in any way beyond what its developers deem appropriate. In others words, the pre-processing and post-processing techniques used to improve output in Google Translate cannot be altered in any way. What this mean is that Google Translate is inappropriate as an academic baseline as it is hard to determine what processing steps were performed on the output beyond SMT. Moses functions as a pure SMT system and allows for customizability and retraining for the purposes of research, allowing researchers to determine which pre-processing or post-processing techniques improve output relative to the pure SMT baseline. It should be noted, however, that Google Translate typically outperforms Moses. Both Moses and Google Translate are used in the current research and will be discussed in Chapter 5 and 6.

2.2.3 Neural Machine Translation

Neural Machine Translation (NMT) is still based in statistical machine learning techniques, however, it is quite a different approach to SMT. In SMT, a translation system typically consists of subcomponents that are seperately optimized, that is, the combination of the highest probability N-grams, while in NMT a large neural network is trained to maximize translation performance (Wolk et al., 2015). NMT models were inspired by representation learning and require only a fraction of the memory used by their SMT counterparts, but each and every component of a NMT system is trained jointly in order to maximize system output (Cho et al., 2014). The complexities of the an NMT system are considerable, and whole papers are written on their explanation, so they will not be discussed in-detail here, as that is not in-line with the objectives of the current research. It is noteworthy, however, that NMTs make use of the most current and cutting-edge technology available in modern MT, and they appear to be leading the charge for the direction of future MT research.

Naver Translate is a freely available NMT system and a comparable counterpart to SMT's Google Translate (2016). Naver Translate traditionally functioned as a SMT system, however, it has recently transitioned to a NMT system, and at the time of writing, is the only known, freely-accessbile NMT system capable of translating Korean to English (Lee et al., 2015). Naver's NMT system makes use of a bi-directional recurrent neural network, the details of which are shown in Figure 2.3:

$$h_t = [\,\overleftarrow{h}_t;\overrightarrow{h}_t\,] \tag{1}$$

$$\overleftarrow{h}_t = f_{GRU}\,(W_{s\_we}\,x_t\,,\ \overleftarrow{h}_{t+1}) \tag{2}$$

$$\overrightarrow{h}_t = f_{GRU}\,(W_{s\_we}x_t\,,\ \overrightarrow{h}_{t-1}) \tag{3}$$

Figure 2.3: Naver Translate's bi-directional recurrent neural network

The details of the system are borrowed from Cho et al., where h_t is a hidden state of the encoder, x_t is a one-hot encoded vector indicating one of the words in the source vocabulary, W_{s_we} is a weight matrix for the word embedding of the source language, and f_{GRU} is a gated recurrent unit (GRU) (2014).[①]

Naver Translate is capable of translating between English, Korean, Chinese, and Japanese, and in that sense is quite limited, however, at the time of writing, Google Translate has not yet trasitioned to a full NMT system for any language pairs beyond Chinese and English (Coldeway, 2016). Though NMT systems generally produce output that is far more desireable than their SMT counterparts, there is still room left for improvement. Additionally, at the time of writing, there has yet to be any research which comments the role of TS for MT as the field transitions to neural networks, which is one of the goals of the present research and will be discussed more in Chapter 6.

2.2.4 Hybrid Machine Translation

Hybrid Machine Translation (HMT) is the combination of two or more MT approaches in one system in order to take advantage of the benefits from both approaches and improve MT system output (Kamran, 2013). This could include any of the MT approaches discussed above or ones not discussed in the present research. For example, the use of a pure SMT Moses system and then the application of a rule-based automatic post-editing system which fixes common errors in SMT output would be considered an HMT system (Rosa, 2013). Such a system would be an example of a statistical-rule-based machine translator. Other combinations are of course possible. The present research, which suggests a rule-

① For more details, please see Bahdanau et al. (2014).

based syntactic simplication system through Korean sentence complexity reduction could be viewed as pre-processing step for an MT system. That being the case, we could view such a system as a rule-based simplfier for a statistically or neural network motivated machine translation system.

2.3 Text Simplification for Machine Translation

As mentioned above, TS used a preprocessing task for NLP tasks, MT in particular, has quite a long history. After Siddharthan introduced his TS architecture, MT motivated rule-based TS systems, not only for English, but for many other language began appearing quite prominently. The main idea being that TS would be helpful from a rule based perspective, as the number of rules and computational difficulty of a sentence would decrease when utilizing TS. If the apporach was statistical in nature, shorter sentences and less complex structures or phrases result in simpler calculations. As Siddharthan, the most prominent figure in the field of automated TS, reminds us, it is simply a fact of language that short, less complex phrasing is more common and therefore easier for a system to learn (2014). Additionally, the longer a sentence gets, the more constrained the environment becomes, making it less likely to see an uncommon or complicated structure or phrasing. In short, TS has become a method by which to compensate for data scarcity.

One on the first pieces of research that explored non-English TS for MT can be found in Specia et al., where researchers developed a rule-based method for the automatic lexical simplification of difficult words in Brazilian Portuguese (2009). Building on her research in 2009, Specia attempted to create an automatic statisical simplifier by creating a parallel corpus of both original and simplified Portuguese sentences and then training them in Moses (2010). This was the first research that attempted to use SMT to translate between simplified and unsimplified text, however, she noted that since Moses learns at a sentence level, this might only work for lexical simplification and not syntactic simplification where a complex sentence is often split into multiple, simpler sentences.

Collados, based on Siddharthan's architecture, developed a Spanish syntactic simplification system and used this system in combination with Google Translate to demonstrate a need for further simplification research done on the Spanish language (2013). His research showed that Spanish TS was able to improve translation quality when translating to disparate languages, such as Korean and Chinese. However, as his was the first TS research conducted on Spanish syntactic simplification, he only dealt with a limited number of structures, namely and-coordination, constrastive-coordination, relative clauses, and cause and effect sentences. Additionally, fearing illegible output, Collados limited his system to only simplifiying sentences that bear two clauses, leaving more complicated structures for future research.

Siddharthan, Collados, Specia, and nearly all TS researchers have noted that one of the greatest issues in automated TS is the lack of an automatic evaluation metric to determine how well a given system is functioning. As Wubben et al., points out, readability metrics, such as the Flesch-Kincaid grade-level metric, should be avoided in TS because such metrics consider only characteristics of the sentence, such as word length, and completely ignore grammaticality or the content's semantic adequacy (2012).

If the goal, then, is MT or some other NLP task, why not use the automatic evaluation metric which is standard to that NLP task? For example, in MT the standard evaluation metric is the BLEU (BiLingual Evaluation Understudy) score (Papineni et al., 2002). The BLEU score essentially compares the GOLD standard translation to the candidate translation by calculating and combining several factors, such as length (modified precision), word choice (recall), word order (distance), while penalizing candidate translations which are significantly longer or shorter than the GOLD standard (brevity penalty(BP)). Please see Figure 2.4 below:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

Figure 2.4: BLEU score calculation

Given the nature of its calculation and considering the nature of TS, the BLEU score is bad metric by which to determine if TS has improved translation quality or not. For example, if we perform a sentence splitting simplifiction where one sentence is separated into two, the translation would reflect that and would most certainly be longer than the GOLD standard. The BLEU score would penalize this in its calculation of length, word order, and brevity penalty leading to a low BLEU score, without reflecting whether or not TS has improved translation quality. The BLEU is also criticized because it has no way of accounting for synonyms. If we perform a lexical simplification and choose a simple word in place of complicated word, then use an MT system, the output would likely reflect that simplification. However, even if the meaning is ultimately the same between the candidate and GOLD standard translations, if a synonym is used in the candidate translation, the BLEU score would penalize us for word choice. As Siddharthan, points out, a low

BLEU score in a TS + MT system may well be indicative of a well functioning TS system (2014).

If readability and NLP automatic evalutation metrics are not a viable option for us, then the only remaining course would be the use of human evaluation. Indeed, unless the explict goal of the work was the exploration of evalutation methods for automated TS, all TS research reviewed for the current research utilized human evaluation to measure the performance of their TS system. However, where previous research focused on evaluting the performace of their TS system in combination with another NLP task, the current research will utilize human evaluation on both the simplified Korean that our system outputs and the translations that our simplifications result in. The reason for this being that the literature addressing the simplification of an Asian language is extremely limited, and therefore provides very little insight on simplification strategies for the task at hand.

2.4 Automated Asian Text Simplification

As stated above, the research addressing the automated simplification of an Asian language is extremely limited. This is a well documented issue in TS, as the majority of the research has been conducted from a European language perspective (Siddharthan, 2014). One goal of the current research is to increase TS's exposure to Asian NLP.

2.4.1 Automated Japanese Text Simplification

The object of the current research is the Korean language, however, the Japanese language enjoys far more attention than Korean on the international stage, and is therefore the object of NLP research more often than Korean. Additionally, though often an item of strong linguistic dabate, it is undeniable that Japanese and Korean bear many similarities in terms of grammatical structure. Therefore, any strategies used in Japanese TS might then be applicable when developing a Korean TS system. However, to date, there has only been one TS study written with a focus on Japanese.

Unfortunately for the present research, Inui et al.'s focus was the use syntactic and lexical paraphrasing for the hearing-impaired whose sign language implements a completely different structure than the Japanese language (2003). As such, the research's focus was not so much fluent output as it was the emphasis of content words and de-emphasis of Japanese grammar not present in Japanese-sign langauge. Given the fact that the researcher's goal was not in line with current research's goal, and the fact that their results were of questionable value, this previous work is of little help to us.

2.4.2 Automated Korean Text Simplification

The Korean language makes use of a phonemic writing system known as Hangeul (한글), which is composed of 14 consonants and 10 vowels. Unlike the English alphabet, Hanguel combines its consonants and vowels together to form a syllabic unit with 11,172 total combinations possible. The combination of these syllabic units form words, though there are indeed many mono-syllabic words in

Korean. For example, 번역 is the combination of four consonants (ㅂ, ㄴ, ㅇ, ㄱ) and two vowels (ㅓ, ㅕ), forming the word *beonyeok* (translation). Until the end of 20th century, Korean was considered to be an Ural-Altaic language as it shares many features with other Ural-Altaic languages, such as the Subject-Object-Verb word order. However, due to the number of borrowings Korean bears from the Chinese language, Korean is currently defined as an isolated language (The National Institute of the Korean Language, 2008).

Considering the disparate nature of Korean and English and the bilingual data scarcity between the two languages, a TS system for an NLP task involving both languages could be very useful. However, as was the case with Japanese TS, there is only one known study that automates Korean TS.

In Chung et al., researchers automate a Korean TS system that improves the readability of Korean newspaper articles on the Internet for the hearing impaired (2013). Their approach is quite interesting as they shift meaningful content to prominent locations in the sentence and make use of the webpage format by highlighting objects of importance. The objective being that Korean-sign language and the Korean language have very different structures, making it difficult for the hearing-impaired to read Korean articles. By emphasizing important content via location and graphical representation, the participants in the study showed a greater reading comprehension when using the TS system. However, it was never the researchers' intention to create fluent Korean output, nor were the researchers attempting to use their output for an NLP task. This previous research is therefore of very little use to the present study.

3. Samsung Machine Translation Corpus

The Samsung Machine Translation Corpus (SC) is a corpus created and provided by Samsung Electronics Limited to the Seoul National University Computational Linguistics Laboratoy for a partner machine translation project. It is the combination of two bilingual corpora of Korean and English, and Korean and Chinese. The origin of the corpus is not entirely certain, however, it contains transcriptions of spoken Korean that is believed to have been subsequently translated into English and Chinese. Though the domain of the corpus meanders in places, the majority of the content in the corpus is based around travel and business. As the primary focus of the current research is Korean and English, the Chinese portion of the corpus will not be discussed in any further detail. There were a total of 354,972 Korean and English sentence pairs.

3.1 Corpus Description

As mentioned above, the SC contains a transcription of spoken Korean, however, unlike many corpora, this corpus is not a collection of conversations, posts, or paragraphs, but rather a collection of isolated sentences that have no bearing on or relation to each other. Additionally, as it is a corpus of spoken Korean, all registers of formality and honorifics are present in the corpus. Korean is an agglutinative language, meaning that morphemes attach to root words to create new word forms and increase word length (Haspelmath et al., 2013). In other words, as the formality and honorific content of the sentence changes, so too does the verb form and verbal ending. Below is an example of the morphological changes the verb 먹다, *meokta* (to eat), undergoes in the present tense as formality register and honorific level changes.

Formality Honorifics	Neutral	Informal	Low-formal	High-formal
No Honorific	먹는다	먹어	먹어요	먹습니다
Honorific	드신다/잡수신다	드셔/잡수셔	드세요/잡수세요	드십니다/잡수십니다

Table 3.1: Example of Korean formality register and honorifics

What this means for the present research is, in order to create a syntactic simplification system that produces fluent Korean output, our system needs to match these verbal endings when reducing sentence complexity by the creation of independent sentences from sentence-internal clauses. This will be discussed in further detail in Chapter 4.

The Korean portion of the corpus bears part-of-speech (POS) tags from what is believed to be an automatic POS tagger, and the POS tags are consistent with the tagging convention described in Han et al. (2001), with the exception of the *ECS* tag which is represented as *EC* in the corpus. The English portion of the corpus also contained POS tags consistent with the Penn Treebank tagging convention (Marcus et al., 1993). However, when the Moses SMT system, used in this study, was being trained, it was discovered that a higher BLEU score was yielded when using only Korean POS tags, and English POS tags were therefore discarded. As a point of reference, the baseline BLEU score yielded by this Moses build was 32.87, a relatively high baseline score for Korean to English SMT. The following table is an example of a bilingual sentence taken from the corpus. Please note that *Korean Tagged* refers to the original sentence in the corpus and is in the same format as the sentences used to train Moses, *Korean Untagged* is the same sentence with tags removed for ease of reading and is in the format used when utilizing other MT systems, like Google and Naver, and *English* refers to the English translation of the original Korean sentence.

Korean Tagged	오늘날/NNG|은/JX 높/VA|은/ETM 급료/NNG|에/JKB 성공/NNG|적/XSN|이/VCP|ㄴ/ETM 직장/NNG|을/JKO 가지/VV|ㄴ/ETM 여성/NNG|들/XSN|을/JKO 보/VV|아도/EC 전혀/MAG 놀랍/VA|지/EC 않/VX|다/EF|./SF
Korean Untagged	오늘날은 높은 급료에 성공적인 직장을 가진 여성들을 봐도 전혀 놀랍지 않다.
English	Today there is nothing surprising about finding a woman holding down a successful job in high finance.

Table 3.2: Corpus sentence example

3.2 Corpus Issues

A number of problems associated with the corpus plagued us during the course of this research. These issues are only addressed here as the thrust of this research is based in human evaluation. Similar problems may have little-to-no bearing on automatic evaluation metrics, but when humans are evaluating output from a system, the input must then also be of high quality.

3.2.1 Korean Issues

During the implementation and development of the simplification system discussed in Chapter 4, a number of mis-tagging issues became apparent during testing of the system. For example, the Korean structure *어/아(서)*, which is a subordinating structure indicating cause and effect or a logical order between clauses, was consistently mis-tagged as follows: 아서/EC, 어서/EC, 아/EC, 어/EC, 아/EC|서/EC, 어/EC|서/EC, 아/EC 아/EC, 어/EC 어/EC, etc. Such inconsistencies had to be accounted for during the implementation our simplification system.

Inconsistencies in tags are not terribly uncommon and part of the struggle in the NLP field, however, one at least expects the language itself to be natural. This was not the case at all as many of the Korean evaluators who participated in this study stated that the original Korean sentences themselves were unnatural or ungrammatical, even before simplification! Such problems may not pose an issue for automatic evaluation metrics, but for human evaluation the quality of the input is critical.

3.2.2 English Issues

The issues plaguing the English side of the corpus were perhaps even worse than the Korean, as the goal of this research is improved translation through simplification. English sentences often bore grammatical errors, for example, if the Korean sentence were "독립문 공원에 갑시다!" the English equivalent was often "Let's go to the Independence Park!" with an out-of-place determiner, *the*. Worse than ungrammaticality, the English sentence often did not bear the same meaning as the Korean. Consider the Korean and English in Table 3.2. The Korean implies that there are many professionally and financially successful women today, however, the English means that many successful women working in the field of finance. On top of this, it was often the case that content from Korean or English would be missing in the equivalent translations. For instance, consider the following sentence pair extracted from the corpus:

(1) a. 석-박사 학위를 취득할 때까지 4년은 걸릴거라 생각해요.

b. I expect to take 4 years to earn a M.D and a Ph.D.. I hope it doesn't take any longer.

While (1a) and the first sentence in (1b) express roughly the same idea, ignoring the fact that *M.D.* refers to medical doctor while *석사* means Master's degree, the second sentence in (1b) has no Korean equivalent and seemingly appears at random. This is an issue that impacts not only human evaluation, but also automatic evaluation, as there would be no Korean equivalent for the English sentence when Moses attempts word-alignment or when the BLEU score is calculated.

4. Korean Sentence Complexity Reduction System

The development of an all-encompassing Korean syntactic simplification system is a daunting task indeed, especially considering the fact that there exists no previous research on which to base such a system. The present research, then, attempts to introduce TS into Korean by performing the most basic syntactic simplification possible: the rephrasing of complex Korean sentences into less complex alternative(s) which bear the same meaning as the original sentence. Recall that sentence complexity refers to the number of clauses in a given sentence. For example, consider (1) below.

(1) a. 오바마가 케냐에서 왔으면 트럼프는 화성에서 왔습니다.

Translation: If Obama's from Kenya, then Trump's from Mars.

b. 오바마가 케냐에서 왔습니다. 그러면 트럼프는 화성에서 왔습니다.

Translation: Obama's from Kenya. Then, Trump is from Mars.

(1a) is an example of Korean sentence internal clause-subordination via a if/then conjunction. As it bears two complete verb phrases, the sentence complexity can then be understood as two. (1b) conveys roughly the same meaning as (1a) and is given logical cohesion via a sentence connector. However this meaning is conveyed in two simpler sentences, each with a sentence complexity of one.

This system is accomplished by the creation of rules which convert Korean clause connecting structures into independent sentences. After successfully creating independent sentences from sentence-internal clauses, addition steps were implemented, such as the insertion of appropriate sentence connectors, in order establish a logic between the sentences and generate a cohesive unit. The ultimate goal for the output of this system is the improvement of MT system output, however, we also hope to generate relatively natural Korean sentences. Additionally, through this system, we suggest a method by which to overcome the initial hurdle of simplifying sentences containing more than two clauses.

4.1 Rule Creation and Description

The Samsung Corpus (SC), which is discussed in Chapter 3, was reviewed and examined for appropriate clause connecting conjunctions. After extracting dozens of sentences from the corpus, several multiple sentence-splitting rewrites were attempted by hand and then translated via Google Translate. If the rewrite was thought to be relatively natural Korean, which also generated a better translation than the original sentence, the manual rewrite was automated via the programming language Java. It should be noted that this is a supervised task, making use of POS tags in order to distinguish between cases of ambiguity caused by Korean homophony. What follows is a detailed description of the rule-based Korean sentence complexity reduction system proposed by the present study.

4.1.1 Korean Coordination Simplification

The most common form of Korean coordination between two clauses would be conjunctive 고 *go* (and), though there are of course other examples. The current research simplified 고-coordinated sentences by inserting a period in place of go 고, or 고/EC as it appears in the corpus, and inserting *그리고* at the beginning of the newly formed next sentence. This operation is applicable to noun, adjective, or verb-ending clauses, though the form changes slightly depending on the word-type in question. For example, please consider the summary below:

(1) a. Clause1 N/A/V-고 Clause2.

Original: 개스킷을 ***교체하고*** 팬벨트가 낡아서 새로운 것으로 교체했습니다.

Translation: (I) replaced the gasket, and since the fan belt was old, (I) replaced it with a new one.

Google Translate: Replace the gasket and replaced by a new fan belt narkahseo.

b. Sentence1 N/A/V-다. 그리고 Sentence2.

Simplified: 개스킷을 ***교체합니다. 그리고*** 팬벨트가 낡아서 새로운 것으로 교체했습니다.

Translation: (I) replace the gasket. And then, since the fan belt was old, (I) replaced it with a new one.

Google Translate: Replace the gasket. And the fan belt narkahseo replaced it with a new one.

Despite the simplicity of the operation performed in (1b), the insertion of a sentence ending morpheme, a period, and a sentence connector, we see a vast improvement in Google Translate's handling of the two newly generated less complex sentences compared to the syntactically complex, original sentence. This is consistent with what we would expect from research conducted in the literature; less complex and shorter sentences are handled more effectively by SMT systems than complex, longer sentences.

There are a few decisions made during the implementation of our complexity reduction system that should be addressed, first among them being tense matching between the reduced sentences. Initially, it was our plan to match tenses between the sentences generated from clauses, for example, instead of generating the simplified sentences in (1b), we would generate the following: 개스킷을 ***교체했습니다. 그리고*** 팬벨트가 낡아서 새로운 것으로 교체했습니다, based on the fact that the sentence as a whole is in the past tense. However, after attempting translation on several such sentences it was found that maintaining the tense the clauses themselves bore generated translations of greater quality. Additionally, not doing so often resulted in over-fitting and forced a meaning on the Korean sentence that was not intended by the original sentence.

Next, one could argue that the insertion of *그리고, kurigo* (and then), increases the complexity of the second clause sentence, therefore, negating our efforts and producing one sentence with a complexity of one followed by a second with a complexity of two. This is due to the fact that, syntactically, one could consider *그리고* as a sentence internal clause connector of the form *그리, kuri* (that way), + *고*, *go* (and).

However, consider the way an MT system sees *그리고* compared to *고*. The context of *고* is a fluid environment, ever shifting between verbs, adjectives, and nouns, and such a system can only determine the meaning of the morpheme *고*

based on frequency of occurrence. Yet, the form and environment it appears in is inconsistent, as it is a productive morpheme. Additionally, the system must then attempt to determine which word 고 best corresponds to in English, even though 고 is not limited to only the translation of *and* in English. 고 is a part of many grammatical structures in Korean, not only conjunctive 고. For example, The morpheme also expresses desire, ~고 싶다, or appears in continuous tense, ~고 있다, etc.

Given these complications, it is little wonder that even a structure as simple and common as 고 poses problems for NLP tasks. On the other hand, consider 그리고, a nonproductive grouping of morphemes with only one form that is nearly always translated as *and then* or a close equivalent. Therefore, from a syntactic perspective, the insertion of 그리고 may be suspect, but from a computational perspective, it clearly improves translation quality by acting as a disambiguation tool. On top of that, in the Samsung Corpus (SC), 그리고 is represented as 그리고/MAG (bearing the MAG POS tag, being a magnifying phrase) and 고 is represented as 고/EC. That being the case, a supervised NLP system, such as the Moses SMT system used in the present research, would not even consider the two words/morphemes to be related because they bear completely different POS tags.

Finally, just as important as a simplification system knowing which structures to simplify, so too must it know when to avoid simplification. As was mentioned above, 고 is an extremely productive morpheme in Korean, appearing in many Korean grammatical structures that express conjunction, desire, continuous tense, etc. However, the simplification system proposed by the present research views it as inappropriate and detrimental to simplify anything beyond conjunctive 고. Therefore, not only for 고, but for every structure handled by this system, exception lists were constructed to determine in which contexts clause reduction should be performed and which contexts it should not. This was achieved by viewing the sentence in question a linear string, finding the morpheme(s) in question, in this case conjunctive 고, and examining the contexts to the left and right of the morpheme as it appears in the string for exceptions. This process is summarized in the pseudocode in Algorithm 1 below.

Algorithm 1 Simplification Exception Check

morph : the conjunction acting as the site of complexity reduction
left : list of structures before the morpheme which form an exception
right : list structures after the morpheme which form an exception
sent : the sentence to be simplified
for all *word* in *sent* **do**
  if *word* contains *morph* **then**
    if *morph* starts with *left* **or**
    if *morph* ends with *right*
      break
    else
      continue
  end if
end for

Though the complexity reduction operation description above is not an exhaustive list of all conjoining structures handled by this system, the vast majority of them operate in a similar manner, following a similar logic by which we avoid the simplification of exceptions. Therefore, for space considerations, examples of simplification via positive coordinating structures will not follow this section, however, a list of structures considered to be coordination will be discussed in section 4.4.

4.1.2 Contrastive Coordination Simplification

Contrastive coordination occurs when two or more clauses are somehow in conflict with each other, and this is realized in English typically in the form of *but*. The proposed complexity reduction system handles contrastive coordination in essentially the same way as the positive coordination example was reduced; the only change being the insertion of a different sentence connecting unit. Please see the example in (2) below:

(2) a. Clause1 N/A/V-은데 Clause2.

Original: 2개월 머물 ***것인데*** 짐 가방이 왜 그렇게 작습니까?

Translation: (We're) staying for two month, but why is (your) bag so small?

Google Translate: 2 months geotinde luggage Why did I stay so small?

b. Sentence1 N/A/V-다. 그런데 Sentence2.

Simplified: 2개월 ***머물것입니다. 그런데*** 가방이 왜 그렇게 작습니까?

Translation: (We're) staying for 2 months. But why is (your) bag so small?

Google Translate: It will stay for 2 months. But luggage is so small, why?

The only difference between how the proposed system reduces sentence complexity for positive and contrastive coordination is the sentence connector inserted, which is determined based on the clause conjoining structure in question. As we can see in the above example, just as we saw with positive coordination in

4.1.1, even the simple operation performed here greatly increases Google translation quality between (2a) and (2b).

Additionally, as was the case with conjunctive 고, Korean homophony did not pose an issue here, because the majority of conjunctive structures targeted by our simplification system bear the ~/EC POS tag, while homophonous structures bear completely different tags. For example, consider 는데/EC vs. 는데/NNB. The former can be translated as *but* in English, and is therefore a target for conjunctive complexity reduction operations, while the latter is often understood as English *while*, and while it is a potential site for clause reduction, the means by which it would be reduced are completely different. However, as our system disambiguates based on POS tags, it does not pose an issue for us.

A list of all contrastive conjunctive structures processed by the proposed system will be introduced in section 4.4.

4.1.3 Indirect Sentence Reduction

Another sentence type the proposed system attempts to reduce are sentences that contain indirect speech or thoughts conveyed via the 다고/EC structure in Korean. Please consider (3) below.

(3) a. TOPIC는 Clause1 N/A/V-다고 Clause2.

Original: ***과학자들은*** 그 열대 우림에 아직 발전되지 않은 많은 종들이 ***있다고 믿고 있다.***

Translation: Scientists believe there many species that have not yet developed in that tropical rainforest.

Naver Translate: Scientists have not yet developed in the rainforest believe that there are many species.

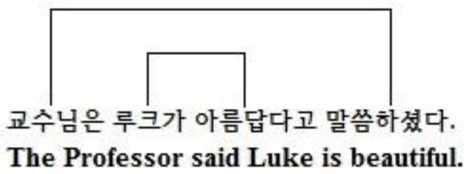
b. Sentence1 N/A/V-다. TOPIC는 그렇게 Sentence2.

Simplified: 그 열대 우림에 아직 발전되지 않은 많은 종들이 있다. ***과학자들은 그렇게 믿고 있다.***

Translation: There are many species that have not yet developed in that tropical rainforest. That's what scientists believe.

Naver Translate: There are a lot of species not yet developed in the rainforest. Scientists believe so.

While similar to the previous reduction strategies introduced so far, indirect sentence simplification varies slightly as we assume the leftmost topic bearing the 은/는/JX morpheme acts the subject of the matrix clause, while subjects or topics, right of the leftmost topic, act as the subject of the embedded clause. Please consider the visualization below:



By assuming this pairing, we therefore copy and track the leftmost topic before splitting the sentence and inserting a period and *그렇게*, *kurutke* (in that way), to establish cohesive logic between the generated sentences. Due to the linear nature of this Korean grammar pattern, where the matrix clause is the rightmost clause, we then are able to simply insert the leftmost topic into the rightmost newly generated sentence, and then delete it from its original position.

While slightly more complicated and syntactically motivated than the previous complexity reductions, this relatively simple operation generates a much better translation using Naver Translate, a cutting-edge NMT system, as can be seen in (3b). A full list of the indirect structures covered by this reduction system will be introduced section 4.4.

4.1.4 Gerund Reduction

Another syntactically complex structure reduced in similar way to the indirect speech structure would be that of Korean gerunds; specifically gerunds of the form verb/adjective~기 acting as the topic, subject, or object of the sentence. Please consider the example in (4) below.

(4) a. TOPIC는 Clause1 A/V-기를 Clause2.

Original: ***부모님은*** 내가 늦게 까지 자지 않고 ***있기를*** 원하지 않으셨다.

Translation: My parents did not want me staying up late.

Google Translate: My parents did not want to be without me stay up late.

b. Sentence1 N/A/V-다. TOPIC는 그것을 Sentence2.

Simplified: 내가 늦게까지 자지 않고 있었다. ***부모님은 그것을***

원하지 않으셨다.

Translation: I did not sleep until late. My parents did not want it.

Google Translate: I do not sleep late. My parents did not want it.

The example in (4) proceeds with a logic very similar to the example in (3), where the leftmost topic is assumed to be the subject of the matrix clause and is therefore copied, inserted into the rightmost sentence after its generation, and then deleted from its original position. When the gerund phrase is the topic or subject of the sentence, such topic reorientation would be unnecessary. Additionally, as this is an example of the gerund functioning as the object of the sentence, the pronoun *그것, kugeut* (that), is inserted to function as the object of the rightmost sentence after the deletion of the gerund phrase. Though the logic is similar, different pronouns are inserted when processing a gerund phrase functioning as the subject or topic of the sentence.

As a comparison between (4a) and (4b) shows, sentence complexity reduction through the creation of independent sentences greatly increases Google translation quality. This is due to the reduced sentence complexity each sentence individually bears, making for much easier processing by the SMT Google Translation system. Additionally, we have substituted a productive grammar structure, with many different possible forms (V/A~기), for a common non-productive pronoun with only one form (그것). In this way we have maintained the meaning of the original sentence through logical cohesion, reduced sentence complexity, and generated better SMT output.

4.1.5 Cause and Effect Reduction

Korean is well known for possessing an uncommonly large number of subordinated structures that express a cause and effect relationship between clauses. All structures bear their own nuanced meaning, however, in English, they are often translated as *because*. Due to this potential many-to-one mapping when translating between languages, the current research proposes a generalized approach for the handling of this sentence type. Given that the nuanced meaning these structures bear will likely be lost in translation anyway, we propose processing these nuance structures in exactly the same way, regardless of their form. Please see the example in (5) below.

(5) a. Clause1 N/V/A-니(까), 해서, 때문/덕분, 인하여(etc) Clause2.

Original: 영어가 ***서툴러*** 영어로 설명할 수 없습니다.

Translation: Because (my) English is clumsy, (I) can't explain in English.

Google Translate: English can not be described as clumsy English.

b. Clause2 → Sentence1. 왜냐하면 Clause1 → Sentence2때문이다.

Simplified: 영어로 설명할 수 없습니다. ***왜냐하면*** 영어가 ***서투르기 때문입니다.***

Translation: (I) cannot explain in English. This is because (my) English is clumsy.

Google Translate: I can not explain in English. This is because English is awkward.

The method by which the current research simplifies cause and effect sentence constructions in Korean is slightly more complicated than our previous simplification proposals. After identifying the structure in question, we split the sentence at the structure boundary with a period, delete the structure (in this case 어/아(서)), and insert 기 때문이다, *gi ddaemunida* (because of), in the original structure's position. We then perform a sentence reordering, where the clause to the left of the simplification site (now a sentence) and the clause to right of the site (also now a sentence) change places. The final step is the insertion of 왜냐하면, *waenyahamyun* (because of), which happens at the beginning of the newly formed sentence containing 기 때문이다. This serves as a means of generating more natural Korean output and as a disambiguating sentence connector to establish a logical flow between the sentences. This process is summarized in the pseudocode in Algorithm 2 below.

Algorithm 2 Cause and Effect Reduction

sent : the sentence(s) being simplified
morph : the morpheme(s) acting as the site of simplification
morph left : the clause/sentence to the left of the simplification site
morph right: the clause/sentence to the left of the simplification site
temp : a placeholder string
for all *words* in *sent* **do**
  if *word* contains *morph*
    morph → "기 때문이다."
    morph left → *temp*
    morph right → *morph left*
    temp → *morph right*
    insert "왜냐하면" at $temp_{-1}$
    return *morph left* + *temp*
    temp → null
  end if
end for

Though slightly more complicated than our initial simplification proposals, the method by which the proposed clause reduction system simplifies sentences is still relatively straight forward. Additionally, it suggests a generalized approach for processing structures that bear nuanced meaning in the source language, but lose this nuance when translated into the target language. Generalizing structures to a more common form or simplification has been a common thread throughout this research and will be discussed in greater detail in section 4.3. On top of this, as can be clearly seen in (5a) and (5b), translation quality is greatly improved when utilizing the proposed sentence complexity reduction approach. A full list of structures processed in this way will be introduced in section 4.4.

Due to space constraints, additional sentence complexity reduction rules utilized in this research will not be discussed any further, however, the most complicated examples have already been introduced and the remaining simplification types follow a similar logic.

4.2 Factorial Complexity Reduction

As was noted in Chapter 2, sentence complexity can pose a substantial issue for syntactic simplification systems when they are first introduced into a language. For example, when Collados introduced his Spanish syntactic simplification system, he was limited to only the simplification of Spanish sentences bearing two clauses, that is, a sentences with a complexity of two (2013). In other words, Collados only performed one simplification per sentence. Considering the pro-drop nature of Korean, where entire clauses or even sentences may be nothing but bare verb phrases with no subjects, objects, or topics, the current research views this hurdle as extremely limiting.

Additionally, when attempting to simplify Korean sentences with a complexity greater than two, there is no basis on which to determine which simplification combination or order would generate the most natural Korean or the best translation. It is possible to force such a decision, however, it is our belief that such a determination should be left up to Korean native speakers evaluating simplified Korean output, and English native speakers evaluating MT output. Therefore, in the present research we propose a method to overcome this initial hurdle and provide foundational insight into what simplifications and simplification combinations generate the most natural Korean and/or the best translations, which we term factorial complexity reduction.

By viewing complex sentences as a string of clause boundaries and making note of these clause boundaries, we are able to view the sentence as a series of clause reduction operations that, when combined together one-by-one, become a cohesive unit of simplified sentences. In other words, we view each simplification possible in a sentence as a factor that, when combined together with other possible simplifications, generate the final, completely simplified unit, hence the factorial nature. We then are able to generate all possible simplifications and all possible simplification permutations in a given syntactically complex sentence. Please see the example in (6) below.

(6) a. 아 죄송하지만 예약을 재확인하지 않으셨기 때문에 성함이 탑승자 명단에서 빠져 있습니다.

Translation: Ah, I'm sorry, but because you did not reconfirm your reservation, your name is not on the passenger list.

b. 아 ***죄송합니다. 하지만*** 예약을 재확인하지 않으셨기 때문에 성함이 탑승자 명단에서 빠져 있습니다.

Translation: Ah, I'm sorry. But because you did not reconfirm your reservation, your name is not on the passenger list.

c. 아 죄송하지만 ***성함이 탑승자 명단에서 빠져 있습니다. 왜냐하면 예약을 재확인하지 않으셨기 때문입니다.***

Translation: Ah, I'm sorry, but your name is not on the passenger list. This is because you did not reconfirm your reservation.

d. ***아 죄송합니다. 하지만 성함이 탑승자 명단에서 빠져 있습니다. 왜냐하면 예약을 재확인하지 않으셨기 때문입니다.***

Translation: Ah, I'm sorry. But your name is not on the passenger list. This is because you did not reconfirm your reservation.

(6a) above is the original sentence before simplification. By scanning the sentence from left to right, we are able to note clause boundaries, determine which simplifications are possible, and then begin clause reduction operations from left to right as they appear in the sentence. (6b) shows the first simplification our system would perform, which is a contrastive clause coordination similar to the example described in section 4.1.2. Our system outputs the simplified sentences, updates the list of factorial simplifications, and continues. The next simplification possible can be seen in (6c) and is of the cause and effect variety, which is described in section 4.1.5. By noting the clause boundaries and only reordering the relevant clauses, the system successfully performs sentence complexity reduction, updates the list of factorial simplifications, outputs the reduced sentences, and continues on. We have now exhausted the number of possible simplifications available in our original sentence, so the only option left is to output the combination of both complexity reduction operations, which can be seen in (6d). For the sake of space and simplicity, we have described an example with a sentence complexity of three, therefore, outputting the two possible syntactic simplifications and one simplification combination. However, there is no limit to the number of complexity reduction operations and permutations our system could perform, so long as the sentence in question were sufficiently complex.

4.3 Phrase Grouping and Generalization

Throughout section 4.1 above, during the descriptions of various rules, we made mention of grouping certain nuance structures together and simplifying them in a manner to motivate disambiguation. For example, in 4.1.5 we discuss the insertion of *왜냐하면* and *때문이다*, which, when translated into English, both roughly equate to *because of*, into the clause bearing the cause of the preceding clause which bears the effect. Following this logic, this research suggests the grouping of infrequent nuance structures together with more frequent structures that carry essentially the same meaning, especially when losing their nuanced meaning during translation into English.

For example, consider the case of 더라도/EC, *duhrado*, and 도/EC, *do*. The former occurs 78 times in the SC while the latter occurs 4,538 times, yet when translated into English, both structures are typically understood as *even though* or *even if*. When we perform sentence splitting complexity reduction operations on these structures, instead of viewing them as unique, this research suggests grouping such clause conjunctions together, generalizing, and performing the same generalized simplification on both. Generalizing to a more frequent form, which the MT system in question has seen more often, increases translation quality as a greater frequency of occurrence is beneficial to all statistical machine learning techniques. For example please consider (7) below.

(7) a. Original: 3 개월 이상에 해당하는 양의 약은 아무리 복용용이라 하더라도 반입하실 수 없습니다.

Translation: Even if you say will take it, you cannot bring more than 3 months worth of medicine.

Naver Translate: Embargo of corresponding amount of medicine is however one for take inmore than three months can not.

b. Simplified: 3 개월 이상에 해당하는 양의 약은 아무리 복용용이라 합니다. 그래도 반입하실 수 없습니다.

Naver translate: Corresponding amount of medicine is no matter how one for taking on more than three months. Can not still carrying them into.

Additionally, when possible, we attempt to substitute frequent, non-productive pronouns or magnifying phrases in place of productive grammar structures that can appear in many different shapes and forms. The reason for this being non-productive words typically only have one form, and are therefore much easier to find a one-to-one translation in the target language than productive conjunctive structures with forms as various as the number of verbs, nouns, and adjectives in a source language. We believe this phrase grouping and generalization logic is not limited to only Korean, and could have an application in all languages as a tool for compensating for data scarcity during NLP tasks.

A list of the phrase groupings made during implementation of this sentence complexity reduction system will be introduced in the next section, 4.4.

4.4 System Coverage

This section details the coverage of the proposed Korean sentence complexity reduction system as it applies to the Samsung Corpus (SC). In other words, how many sentences within the corpus could our system apply to and simplify? Technically, as the system was constructed with a filter to sort out sentences with insufficient complexity, that is, sentences bearing only one verb phrase, our system has 100% coverage as it would ignore sentences with insufficient complexity or sentences that are only comprised of expectations. As was mentioned in section 4.1.1, not simplifying inappropriately is just as important as successful simplification. However, disregarding this logic, the following table summarizes the grouping of conjunctive structures our system applies to and the percentage of sentences in the corpus they appear in. As a point of reference, there are a total of 354,972 sentences in the corpus. Structures are divided by parenthesis and commas for ease of viewing.[②]

② As was mentioned in section 4.1.1, when the proposed system checks for simplification exceptions, it looks to the right and left of the simplification site. This is reflected in grouping marked with *exception*, as the conjunctive structure may have a *null* to its immediate right or left. This simply means the system need not check to the right or left of the simplification site.

<table>
<tr><th>Grouping</th><th>Conjunctive Structures</th><th>Percentage of Application</th></tr>
<tr><td>And</td><td>(|고/EC),(|고/EC나/VX|서/EC),(|고/EC|는/JX),(고/EC|,/SP),(|고/EC나/VX|서/EC|,/SP),(|고/EC|는/JX|,/SP)</td><td>7.89 %</td></tr>
<tr><td>And Exception</td><td>(null, " 나/VX"),(null, " 말/VX"),(null," 계셔/VX"),(null," 계시/VX"),(null," 싶/VX"),(null," 있/VX"),("타/VV"," 다니/VV"),("알/VV"," 보/VV"),("알/VV"," 보/VX"),("모시/VV"," 가/"),("모시/VV"," 오/"),("놓/VV"," 오/"),("놓/VV"," 가/"),("갖/VV"," 가/"),("갖/VV"," 오/"),("데리/VV"," 오/"),("데리/VV"," 가/"),("무릅쓰/VV", null), ("/EC 가지/VX", null)</td><td>10.71%</td></tr>
<tr><td>Reason</td><td>(|아서/EC),(|어서/EC),(|아/EC),(|어/EC),(|라서/EC),(|는/ETM 바람/NNB|에/JKB),(|어/EC 가지/VX|고/EC),(|아/EC 가지/VX|고/EC),(|느라/EC),(|느라고/EC),(인하/VV|아/EC),(인하/VV|아/EC|서/EC),(|으므로/EC),(|므로/EC),(|기/ETN 때문/NNB|에/JKB),(|때문/NNB),(|으니/EC),(|니/EC),(|으니까/EC),(|니까/EC),(|니까/EC),(ㄴ/ETM 덕분/NNG),(ㄴ/ETM 탓/NNG)</td><td>22.61%</td></tr>
<tr><td>Reason Exception</td><td>(null, " 두/"),(null,"|야/"),(" 대하/VV", null),("|에/JKB|대하/VV", null),("|에/JKB 대하/VV", null),(" 위하/VV", null),("|에/JKB 위하/VV", null),("걸/VV", " 오/"),(" 걸/VV", " 가/"),(null," 싶/VX"),(null," 있/"),(null," 주/"),(null," 드리/"),(null," 계셔/VX"),(null," 계시/VX")</td><td>21.95%</td></tr>
<tr><td>Even</td><td>(|더라도/EC),(|아도/EC),(|어도/EC),(|어/EC|도/EC),(|아/EC|도/EC),(|라도/EC)</td><td>1.3%</td></tr>
<tr><td>Even Exception</td><td>(null, " 되/"),(null, " 좋/"),(null, " 괜찮/"), (null, " 있/")</td><td>.95%</td></tr>
<tr><td>If/Then</td><td>(|면/EC),(|으면/EC),(|ㄴ다면/EC),(|다면/EC),(|는다면/EC),(다면/EC|,/SP),(|면/EC|,/SP),(|으면/EC|,/SP),(|ㄴ다면/EC|,/SP),(|다면/EC|,/SP),(|는다면/EC|,/SP)</td><td>3.89%</td></tr>
<tr><td>If/Then Exception</td><td>("그렇/VA", null),(null, " 되/"),(null, " 좋/")</td><td>1.4%</td></tr>
<tr><td>But1</td><td>(지만/EC),(지만/EC|,/SP),(습니다/EC|만/JX|,/SP),(습니다/EC|만/JX)</td><td>1.21%</td></tr>
<tr><td>But2</td><td>(ㄴ데/EC),(는데/EC),(은데/EC),(ㄴ/ETM|데요/</td><td>1.28%</td></tr>
</table>

	EC),(는데요/EC),(은데요/EC),(ㄴ/ETM|데요/EC|,/SP),(데요/EC|,/SP)	
While	(/ETM 데/NNB),(|ㄹ/ETM 때/NNG),(|ㄹ/ETM 때쯤/NNG),(|을/ETM 때/NNG),(|을/ETM 때쯤/NNG) 5	.9%
Temporal	(|며/EC),(|으며/EC),(|며/EC|,/SP),(|으며/EC|,/SP),(|면서/EC),(|면서/EC|,/SP),(|으면서/EC),(|으면서/EC|,/SP)	.2%
After	(/VX|다가/EC),(ㄴ/ETM 후/NNG),(ㄴ/ETM 후/NNG|에/JKB),(ㄴ/ETM 후/NNG|,/SP),(ㄴ/ETM 후/NNG|에/JKB|,/SP),(ㄴ/ETM 다음/NNG),(ㄴ/ETM 다음/NNG|에/JKB),(ㄴ/ETM 다음/NNG|,/SP),(ㄴ/ETM 다음/NNG|에/JKB|,/SP)	.26%
Before	(기/ETN 전/NNG|에/JKB),(기/ETN 전/NNG),(기/ETN 전/NNG|,/SP),(기/ETN 전/NNG|에/JKB,/SP)	.28%
Gerund	(|기/ETN|를/JKO),(|기/ETN|ㄹ/JKO),(|기/ETN|가/JKS),(|기/ETN|는/JX)	.34%
Indirect	(자고/EC),(냐고/EC),(이/VCP|라고/EC),(라고/EC),(ㄴ다고/EC),(다고/EC),(ㄴ/ETM|다고/EC)	1.5%
Or	(거나/EC),(거나/EC|,/SP),(으나/EC|,/SP),(으나/EC|),(나/EC)	.19%
17 Groupings	137 structures	Total corpus coverage: 76.86%

Table 4.1: System Coverage

Whether it be the application of a simplification or knowing when a simplification would be inappropriate, the proposed complexity reduction system can successfully review approximately 77% the sentences in the SC. This is achieved by grouping 137 structures into 17 different sets, generalizing, and simplifying based on the rules described in section 4.1. This is in no way an exhaustive list, as there are likely grammar structures not considered by the current study. However, we are confident that the system could be simply updated to incorporate new conjunctions should they become known. For example, a conjunction not considered by this research would be that of N/A/V든지, which roughly expresses an equality between clauses, which in English is often realized as *or*. Though untested by the current research, we believe such a structure would easily fall within the *Or* grouping provided above. We leave the discovery of additionally conjunctions to future research.

4.5 System Architecture

The proposed Korean sentence complexity reduction system is heavily based on the simplification architecture introduced by Siddharthan (2002). However, a shortcoming of Siddharthan's research is its exclusive focus on European languages, therefore, some of the steps he suggests are inappropriate for Korean simplification, and the Samsung Corpus (SC) in particular. Therefore we introduce the following architecture for the proposed system in Figure 4.1:

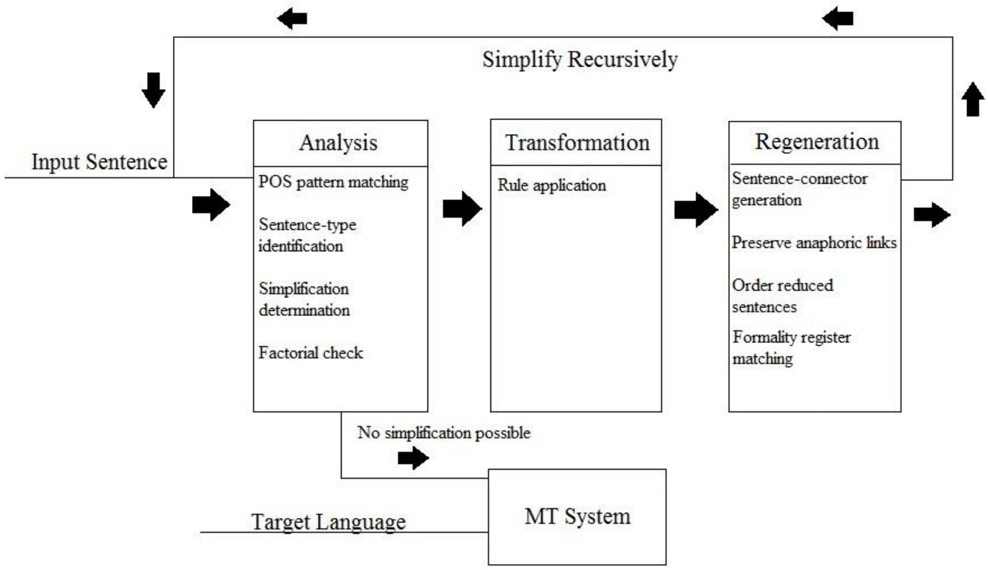


Figure 4.1: Korean Sentence Complexity Reduction Architecture

Using this architecture, we take a Korean input sentence and insert it into our system. In the first stage we perform the necessary analysis such as using POS tags to parse the sentence, identify the sentence type (question, statement, etc) determine what simplifications, if any, to perform, and finally make note of which factorial permutations are possible in the sentence, as described in section 4.2. Based on our findings during analysis, the system determines which rules to apply and does so based on the rules described in section 4.1. The newly generated sentences are then brought to the final stage where we establish a cohesive logic by the insertion of sentence connectors, move anaphors, such as topics if necessary, into their appropriate positions.

In the final stage we also match the formality between sentences generated by the proposed system. As was mentioned in Chapter 3, the SC is a corpus of spoken Korean, therefore, all registers of Korean formality and honorifics are present in the corpus. In order to generate Korean sentences that sound as natural as possible, we thought it necessary to maintain the formality of the original Korean sentence in our reduced sentences. This was accomplished by making lists of all possible formality endings in the corpus, noting the original sentence's formality and type, then simply inserting the appropriate ending before the sentence ending punctuation at the simplification site. Though not a necessary step for Korean to English MT, as English has no equivalent formality registers, this, we believe, is an imperative step for any Korean syntactic TS system attempting to produce natural output.

After the final regeneration stage, the reduced Korean sentences are returned to the initial stage and the process begins anew, in this way simplifying and generating factorial permutations recursively. Should no simplification be possible, the reduced Korean sentences are inserted into an MT system, and then converted into the target language. Though the focus of the current research is Korean to English translation, we believe this system should be able to improve translation output for any MT project where the source language is Korean.

4.6 System Evaluation

As was mentioned in Chapter 1, both Korean and English output from this system will be evaluated by native speakers in Chapter 5 and 6. However, in order to determine if the system is functioning within our specifications, precision, recall, and the f1 score were also calculated.

These evaluation metrics were originally developed for Information Retrieval systems, however, are commonly used to evaluate NLP systems, TS systems included (Schütze, 2008). Precision can be understood as the percent of selected items that are correct, in our case, the percentage of simplified items that should have been simplified. Recall can be understood as the percent of correct items that were selected, and in our case, the percentage of correct items that should be simplified, whether simplified by the system or not.. And finally, the f1 score is simply a cumulative measure of precision and recall.

For the experiment described in Chapter 6, 70 sentences of varying complexity were extracted from the corpus and ran through the proposed system. Precision, recall, and f1 were calculated based on the simplifications performed, or not performed, by our system on these 70 sentences. Please see Table 4.2 below which summarizes the evaluation results.

Metric	Value
Precision (% of selected items that are correct)	94.59%
Recall (% of correct items that are selected)	98.59%
F1 (Cumulative measure)	96.55%

Table 4.2: System Evaluation

According to these metrics, our system is performing quite well with high numbers all across the board. This is the case for two reasons. First, simplification studies typically have high precision and recall because the scale is typically very low due to TS's reliance on human evaluation. For example, in (Collados, 2013), researchers had a precision of 100% and a recall of 92%. Next, the proposed system was implemented for a corpus bearing POS tags, and is therefore a supervised task. It then became a very simple matter to filter out problematic structures based simply on what POS tag the structure bore.

One might then ask why our precision is not 100%, and that is completely due to the problematic Korean structure of *어/아(서)*, *uh/ah*(*suh*). This conjunction can denote two meanings between clauses, either cause and effect or a logical sequence of events. These different meanings entail completely different simplifications by our system. However, in the SC, there is no way to distinguish between these two meanings, as they are consistently tagged with the POS tag */EC*. In order to deal with this problem we attempted to add an exception that said any clause immediately following this structure that started with a verb phrase was to be ignored. However, this too created problems, please consider the example in (8) below.

(8) a. 갑자기 비가 와(서) 들어갔어요.

b. 앉아(서) 먹어.

(8a) is a perfect candidate for cause and effect reduction described in section 4.1.5, however, (8b) is not. If the present research had excluded sentence patterns like (8b), we also would have wrongly excluded (8a). Wanting to still have the option of simplifying sentences like (8a), we decided to permit our system to wrongly simply sentence types like (8b), and generate false positives rather than miss a simplification opportunity. In this way the present research favors recall over precision. It is worth noting the factorial nature of our system described in the section 4.2. As the proposed system generates all possible simplifications and all possible simplification combinations, false positives, we think, are much preferable to missed opportunities, as system output could be easily pruned for unwanted sentences.

5. Pilot Study

With the simplification system described in Chapter 4 in place, a pilot study was conducted to determine the quality of the complexity-reduced Korean sentences generated by our system and whether or not said sentences had a meaningful improvement on MT output.

The sentences in the Samsung Corpus (SC) were randomly shuffled, and the corpus itself was divided into three parts in order to train the Moses SMT system: 90% for training, 5% for tuning, 5% for testing, as is standard in most NLP experiments (Jurasky et al., 2014). The format of the sentences used to train Moses is described in Chapter 3, and the build used in this experiment yielded a baseline BLEU score of 32.87, which is quite high for Korean to English SMT. Both Moses and Google Translate were used in this experiment in order to examine the effectiveness of simplification as a preprocessing task for Korean to English MT in a corpus-sensitive environment, such as Moses, and a non-corpus-sensitive environment, such as Google Translate.

5.1 Methodology

Based on the groupings described in section 4.4, 27 Korean sentences bearing potential simplification site(s) of varying sentence complexity were extracted from the SC. If an extracted sentence contained more than three possible simplifications it was discarded and replaced with a new sentence. This is because, due to the factorial nature of our system, such sentences would generate a large number of subsequent simplified sentences and factorial permutations, and we felt this would likely overwhelm and fatigue our evaluators. These sentences were then input into the proposed Korean complexity reduction system described in Chapter 4. This resulted in a total of 78 sentences: the 27 original sentences and 51 simplified sentences. These sentences were then given to 5 Korean native speakers who were asked to numerically evaluate simplified sentence quality relative to the original sentence in terms of 4 dimensions.

The 4 dimensions used in this study were meaning retention (does it maintain the original meaning), fluency (are the sentences grammatical), naturalness (do the sentences seem like something you might say or hear), and readability (were the sentences difficult to read or understand). These 4 dimensions are commonly used in much of the TS literature (Siddharthan, 2014). Participants were asked to evaluate on a scale of 1 - 10, with 1 indicating low quality and 10 indicating high quality. An example of the simplification format is provided below in Table 5.1.

The original sentence before simplification acts a point of reference for the evaluators. The quality of the original sentences is not a point of interest for this current research and was therefore not evaluated. Please note that there are two potential simplification sites in the example in Table 5.1, namely *있습니다만*, and *등록하기 전에*. *Simplification 1* is an altered version of the original sentence after having undergone one simplification, so that now it has been paraphrased into two sentences, and is therefore evaluated. *Simplification 2* is another paraphrased version of the original sentence, having undergone a different simplification. *Simplification 3* is the combination of these two factorial permutations, that is, both *simplification 1* and *2* combined.

The sentences used in the Korean simplification evaluation experiment described above were then run though Moses and Google Translate, and then paired with their English counterpart in the corpus. This counterpart sentence will be termed the desired translation, as this is the translation the MT systems are attempting to generate. The desired translation was introduced with the MT output in order for the evaluators to understand the intended meaning of the subsequent translation attempts. This resulted in a total of 183 sentences, which were reviewed by 5 English native speakers to determine translation quality. English speakers evaluated based on the same 4 dimensions described above, however, in addition they were asked a yes-or-no question as to whether or not overall translation quality had improved relative to the original translation. To the best of our knowledge, this is the first TS study to include the use of such a metric. Please see Table 5.1 below.

System	Output
Simplification Original	예, 그렇게 생각하고 있습니다만, 대학원에 등록하기 전에 어학 코스를 택하려고 합니다.
Simplification 1	예, 그렇게 생각하고 있습니다. 하지만 대학원에 등록하기 전에 어학 코스를 택하려고 합니다.
Simplification 2	예, 그렇게 생각하고 있습니다만, 대학원에 등록합니다. 그전에 어학 코스를 택하려고 합니다.
Simplification 3	예, 그렇게 생각하고 있습니다. 하지만 대학원에 등록합니다. 그전에 어학 코스를 택하려고 합니다.
Simplification + Moses SMT Desired Translation	Yes, I do. But I will take intensive language course before enrolling in the graduate school.
Moses Original	Yes, so thinking subsidairy, but Registration to graduate school before language course I'd like you choose.
Moses 1	Yes, think so. But Registration to graduate school before language course I'd like you choose.
Moses 2	Yes so thinking subsidairy, but Registration to graduate school. language course I'd like you choose.

Moses 3	Yes think so. But to graduate school Registration. before that language course I'd like you choose.
Simplification + Google SMT Desired Translation	Yes, I do. But I will take intensive language course before enrolling in the graduate school.
Google Original	Yes, but I think so, to choose a language course before enrolling in graduate school
Google 1	Yes, I think so. But I am trying to choose a language course before enrolling in graduate school.
Google 2	Yes, I think so, but must enroll in graduate school. Before that try to choose a language course.
Google 3	Yes, I think so. However, I will register in graduate school. Before that trying to choose a language course.

Table 5.1: Pilot Study format example

Simplification original or numbered is parallel to Moses or Google original or numbered, respectively, though please note that Koreans only evaluated the simplification output, and English speakers only evaluated the Moses and Google output.

5.2 Simplification Results

Once all 5 Korean evaluators returned their evaluations, the means were collected and statistical analysis was performed in the programming language R. First, to get a general idea of the how natural Koreans considered the system output to be, we looked at the means of the four evaluations dimensions summarized in Table 5.2 and Figure 5.1 below.

Dimension	Mean
Fluency	7.88
Meaning Retention	7.84
Naturalness	7.45

Readability	8.33

Table 5.2: Simplification Means

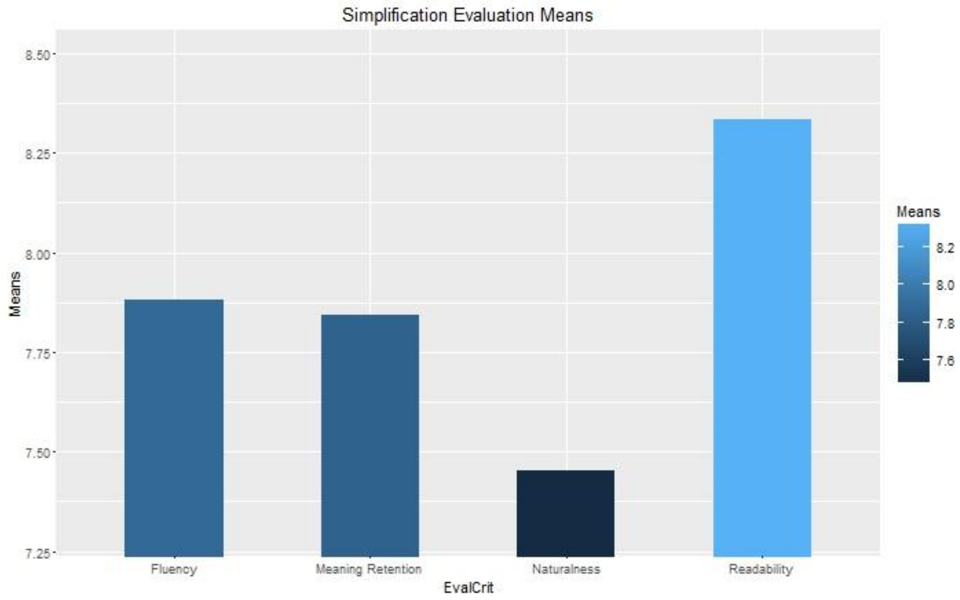


Figure 5.1: Simplification Evaluation Means

In broad stroke summarization statistics, the evaluation of our Korean simplifications seems quite high, as in general only the *naturalness* dimension slightly dips beneath a 7.5. For a more in-depth look, analysis of variance (ANOVA) were calculated, ANOVA being commonly used in TS studies to determine statistically different means between groups. The purpose being to examine if the simplification performed resulted in a significantly different evaluation score compared to other simplifications. The process by which simplifications were grouped and performed in the present study is discussed in Chapter 4.

Comparing simplifications as groups yielded statistically significant results for all dimensions considered at $p = 0.05$, excluding *readability* which was significant at $p = 0.1$. This is consistent with our examination of dimensional means, as *readability* was evaluated substantially higher than the other dimensions. Our findings are summarized in Table 5.3 and the following graphs below. In the figures that follow, the Y-axis represents the simplification type(s) performed and the number of simplifications performed. For example, AND_REASON indicates the combination of a simplification on the conjunctions in the *And* grouping and in the *Reason* grouping, which is detailed in Chapter 4. While the X-axis indicates said grouping's dimensional score.

Dimension	P-value
Fluency	**0.0409**
Meaning Retention	**0.0114**
Naturalness	**0.0239**
Readability	**0.0716**

Table 5.2: Simplification ANOVA by evaluation dimension

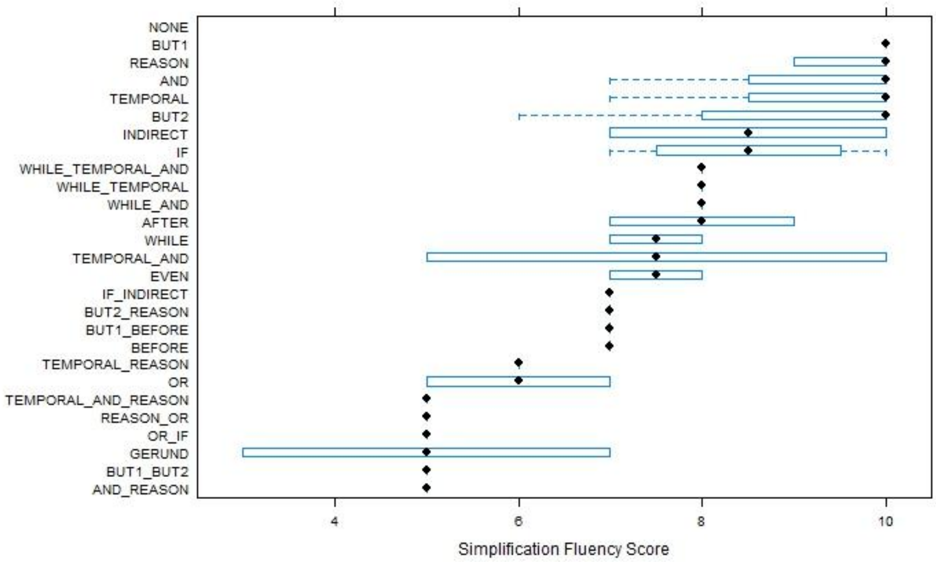


Figure 5.2: Fluency score by simplification performed

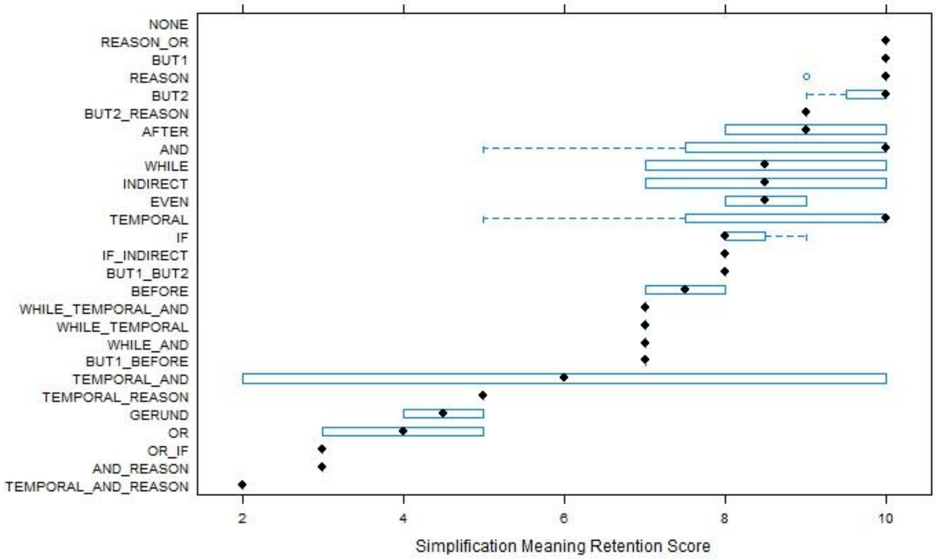


Figure 5.3: Meaning retention score by simplification performed

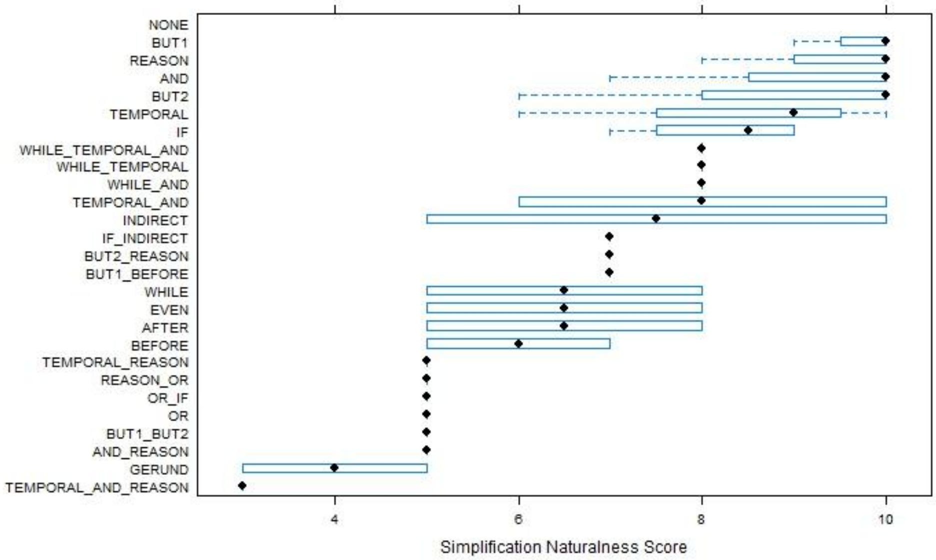


Figure 5.4: Naturalness score by simplification performed

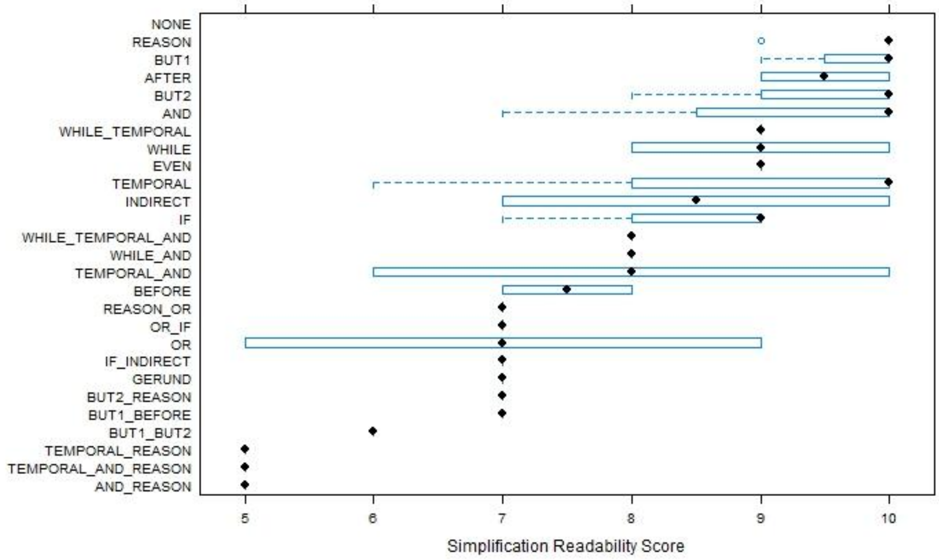


Figure 5.5: Readability score by simplification performed

Though there are indeed a few exceptions, such as the *Or* grouping, in general we see a consistent pattern for each dimensional score. That is, sentences that only underwent one simplification were generally evaluated to be of higher quality than other factorial permutations and combinations, which are discussed in Chapter 4. This is intuitive and not terribly surprising, as the more simplifications that are performed on a given sentence, the more choppy and unnatural the resulting sentences are likely to seem.

5.3 Simplification + Machine Translation Results

Once the English speakers returned their evaluations, the means were again calculated. First, we examined whether or not simplification improved translation quality by the binary grouping of *yes* or *no* in the question of overall translation quality improvement. For Google Translate, the proposed Korean sentence complexity reduction system improved translation quality 71% of the time, while Moses translation quality was improved 78% of the time. Please see Figure 5.6 below.

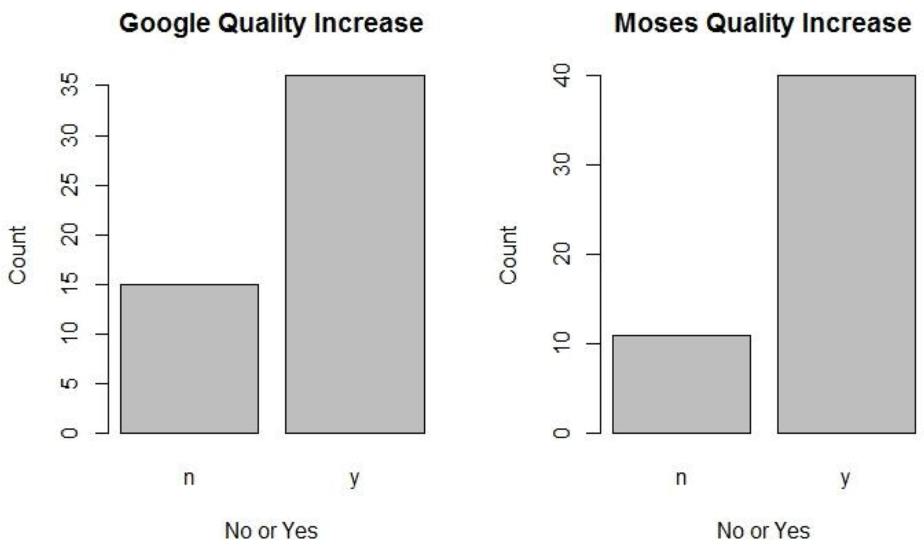


Figure 5.6: Google and Moses translation quality increase

To see this in terms of which reduction operations resulted in better translation quality, we then examined this binary grouping by simplification performed. The pattern we noticed is that, in general, the proposed system does seem to improve translation quality for Korean to English MT. Sentences that did not result in an improvement, tended to be simplification combinations, such as WHILE_TEMPORAL_AND, however, this was inconsistent between Google and Moses. Indeed, while there was great deal of overlap in terms of which simplifications improved Moses and Google output, there is no overlap between Google and Moses for which structures did not improve translation quality. Please see Figure 5.7 and 5.8 below.

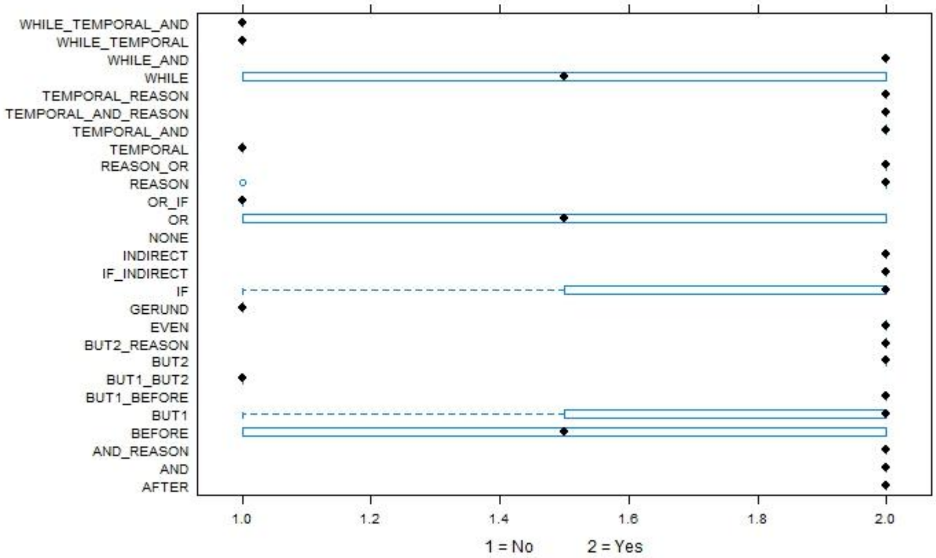


Figure 5.7: Google translation quality increase by structure

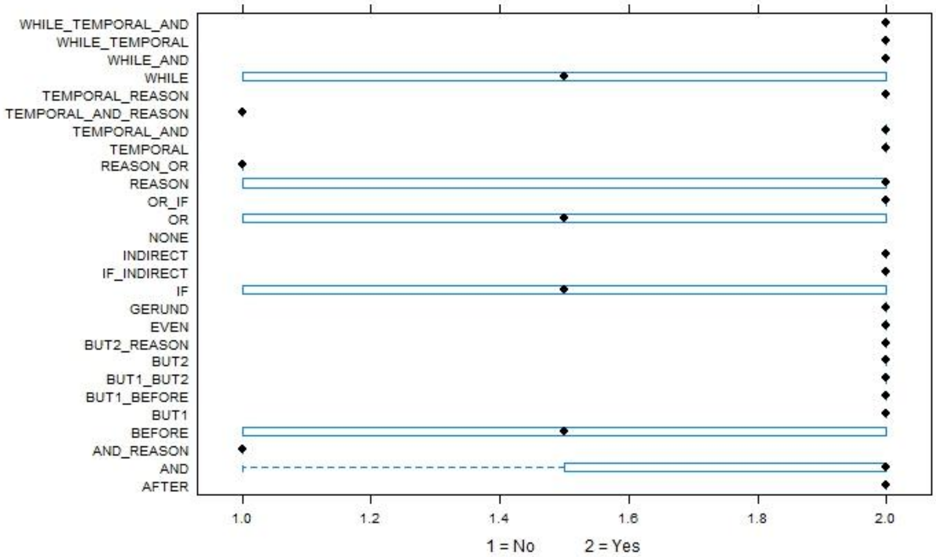


Figure 5.8: Moses translation quality increase by simplification

ANOVAs were then conducted in a similar manner as above, comparing each evaluation dimension against simplification type. However, no statistically significant difference was yielded. This is not particularly surprising. As the graphs above show, single complexity reduction operations nearly always resulted translation quality increase, and since we did not control for the number of simplifications available in a given sentence, these single operations vastly outnumber the multiple simplification operations, which were more likely to not improve translation quality. The ANOVA results are provided Table 5.3 below.

Dimension	P-value
Google Fluency	.695
Google Meaning Retention	.71
Google Naturalness	.715
Google Readability	.88
Moses Fluency	.782
Moses Meaning Retention	.353
Moses Naturalness	.88
Moses Readability	.902

Table 5.3: Translation evaluation dimension by simplification ANOVA results

Next, looking further into the relationship between evaluation dimension and the overall translation quality increase, ANOVAs were executed to determine whether one was predictive of the other. That is, if a given evaluation dimension were to change significantly, would we expect it to have a significant influence on overall translation improvement? Comparing the binary grouping (yes or no) of overall translation quality increase against the four dimensions yielded only one significant result, which was Google meaning retention. Please see Table 5.4 and Figures 5.9 and 5.10 below.

Dimension	P-value
Google Fluency	.275
Google Meaning Retention	**.0307**
Google Naturalness	.579
Google Readability	.181
Moses Fluency	.591
Moses Meaning Retention	.139
Moses Naturalness	.421
Moses Readability	.636

Table 5.4: Translation evaluation dimension by translation quality increase ANOVA results

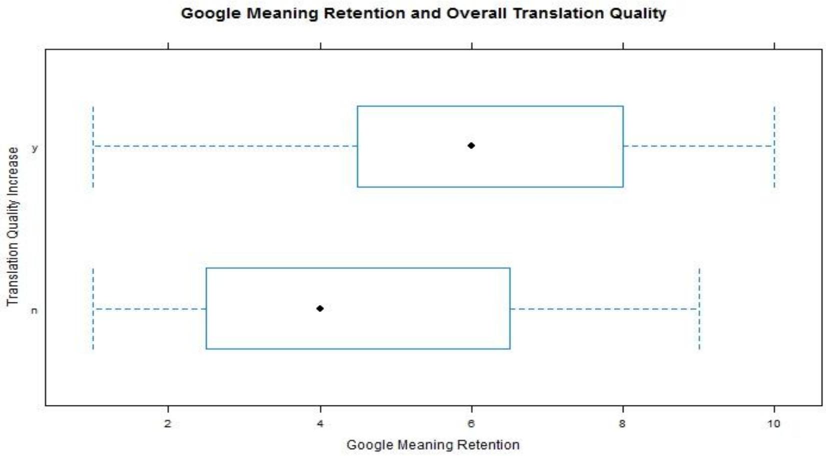


Figure 5.9: Google meaning retention by translation quality increase box-plot

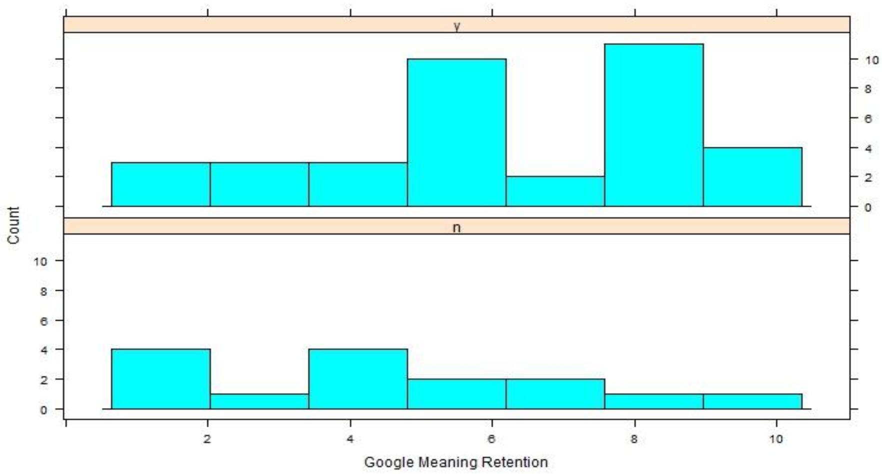


Figure 5.10: Google meaning retention by translation quality increase histogram

As the ANOVA results and graphs above show, Google meaning retention and the translation quality of Google MT output seem to be related. That is, as the meaning retention of a given sentence increases, so too would we expect overall translation quality to increase. This relationship between meaning retention and translation quality gives an indication of how to focus future Korean TS research for machine translation. In other words, conveying the appropriate meaning may increase translation quality more than generating fluent output.

Finally, ANOVAs were performed to compare whether or not the proposed system as a preprocessing task for MT had a significant relationship on any of the four evaluation dimensions. A binary grouping of pure MT vs. SIMP + MT (simplification as a preprocessing task for MT) was created to compare these two types of output against each other. Only one statistically significant difference was captured: that of Moses meaning retention against MT vs. SIMP + MT. ANOVA results are provided in Table 5.5 below, along with accompanying graphs in Figures 5.11 and 5.12.

Dimension	**P-value**
Google Fluency	.678
Google Meaning Retention	.228
Google Naturalness	.949
Google Readability	.872
Moses Fluency	.358
Moses Meaning Retention	**.00151**
Moses Naturalness	.433
Moses Readability	.439

Table 5.5: Translation evaluation dimension compared against MT vs. SIMP+MT ANOVA results

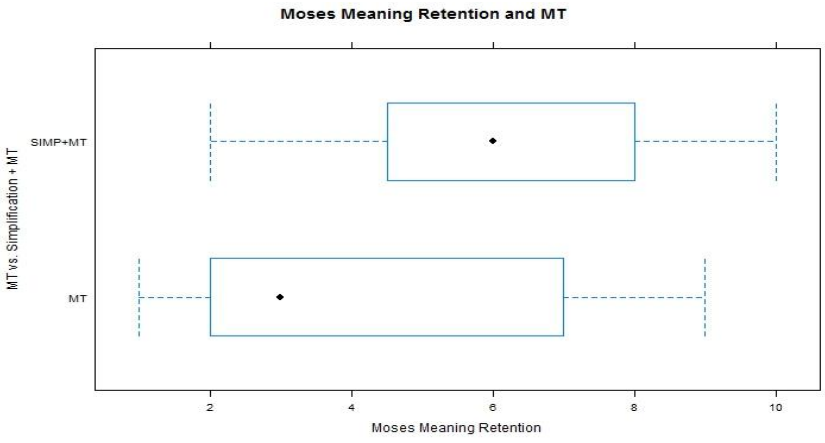


Figure 5.11: Moses meaning retention by MT vs. SIMP + MT box-plot

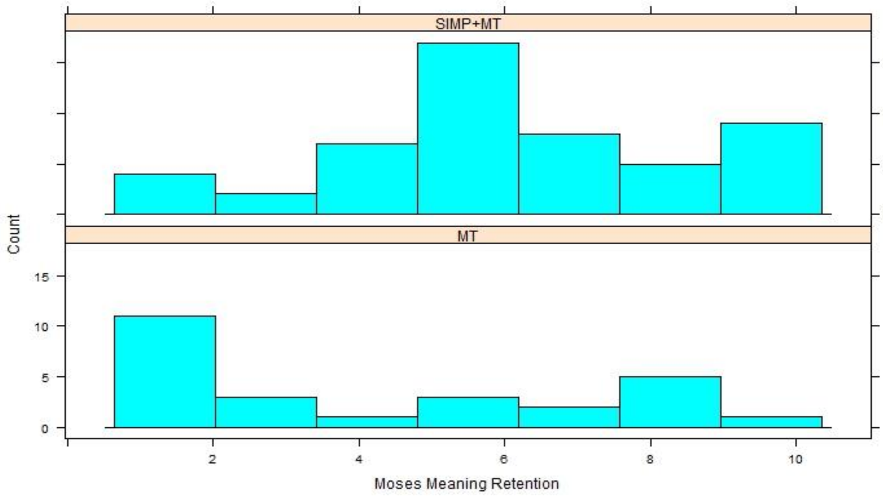


Figure 5.12: Moses meaning retention by MT vs. SIMP + MT histogram

As the ANOVA results and the above graphs indicate, simplification of Korean text and then the subsequent application of a Moses MT system significantly improves Moses meaning retention compared to pure Moses MT. As we can see, when the simplification is applied, the variance is much more compact and results in higher meaning retention scores. When it is not, there is significant left skewing, which on these graphs, is indicative of negative results.

5.4 Pilot Study Discussion

The results captured in section 5.3 were encouraging enough to warrant a larger-scale experiment. The proposed Korean sentence complexity reduction system seems to output relatively high-quality Korean sentences. These simplified sentences, in turn, seem to have a positive impact on both Moses and Google SMT system output. An interesting development to note is the relationship between the number of clause reduction operations applied to a sentence and their effect on sentence quality and translation quality. Please consider Figure 5.13 below. In the graphs that follow, meaning retention was examined, however, all evaluation dimensions yielded similar results.

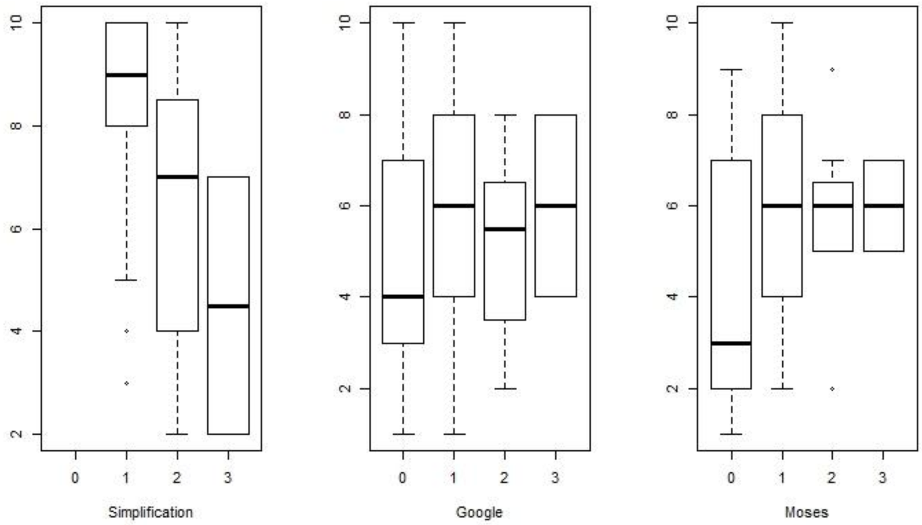


Figure 5.13: Meaning retention score and number of complexity reduction operations

What we see in Figure 5.13 is that as the number of simplifications increase, Korean sentence quality steadily decreases, which is intuitive. The constant breaking and splitting of normal sentences would likely feel choppy and unnatural for a Korean speaker without literacy issues. However, for MT output, we see a dramatic meaning retention increase when one simplification is performed, but subsequent simplifications seem to have no great effect, and instead result in a leveling-out. As was mentioned earlier, simplification number was not controlled for in this study, so we cannot comment on it beyond what we have noticed in our limited exploration here. We will return to this issue in the next chapter, Chapter 6.

6. Experiment

Encouraged by the results yielded in the pilot study described in the Chapter 5, a large scale experiment was conducted. In addition to evaluating MT output from Google Translate and Moses, SMT systems, our English speaking evaluators also evaluated Naver Translate output, an NMT system. To the best of our knowledge, this is the first study to address text simplification as a preprocessing task for neural network motivated machine translation.

6.1 Experiment Methodology

The methodology used in this experiment is largely the same as the methodology described in section 5.1, excluding two differences.

First, the scale of this experiment is much larger than the pilot study, as 70 sentences were extracted from the corpus for simplification and subsequent translation.[③] This resulted in a total of 210 sentences evaluated by 5 Korean native speakers, and 700 translations evaluated by 5 English native speakers. To our knowledge, no TS study has ever been conducted at this scale.

And second, while the evaluation dimensions remained the same, the scale was reduced to 1 - 5, with 1 being low quality and 5 being high quality. This is because many evaluators expressed discomfort with the 1 - 10 scale, claiming it was too unwieldy for evaluation.

③ Before these sentences were extracted, a filter was set to sort out any sentence that appears in the Samsung Corpus more than twice. There are many repeat sentences in the corpus, and if Moses was trained on a sentence it had seen multiple times before and then asked to translate that sentence, it would produce falsely proficient output. This filter should have no effect on Google and Naver Translate.

6.2 Simplification Results

The means of the four dimensions were again calculated to determine, in a broad sense, how natural the Korean evaluators considered the proposed system's output to be. These are included below in Table 6.1 and Figure 6.1.

Dimension	Mean
Fluency	3.08
Meaning Retention	3.7
Naturalness	3.03
Readability	3.4

Table 6.1: Simplification Means

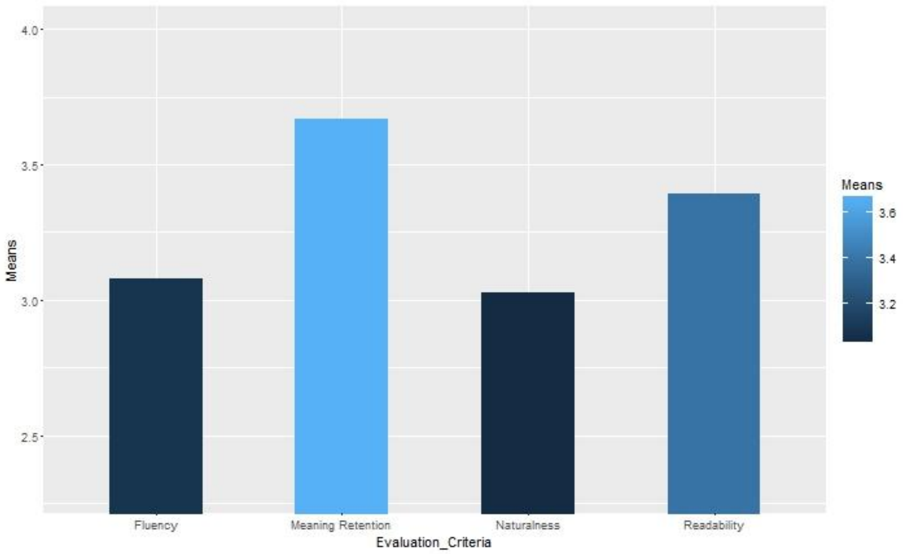


Figure 6.1: Simplification Means

Bearing in mind the different scale used between the pilot study and this experiment, we can see that, while not rated quite as high as in the pilot study, in general our system produces output of relatively high-quality Korean, especially in terms of meaning retention, reaching a mean of 3.7.

In order to look at the significance of complexity reduction operation on Korean sentence quality, ANOVAs were again conducted comparing each simplification type within each evaluation dimension. All four ANOVAs yielded significant results. Please see Table 6.2 and the following graphs.

Dimension	**P-value**
Fluency	**0.000129**
Meaning Retention	**9.49e-07**
Naturalness	**2.3e-05**
Readability	**4.39e-06**

Table 6.2: Simplification type by evaluation dimension ANOVA results

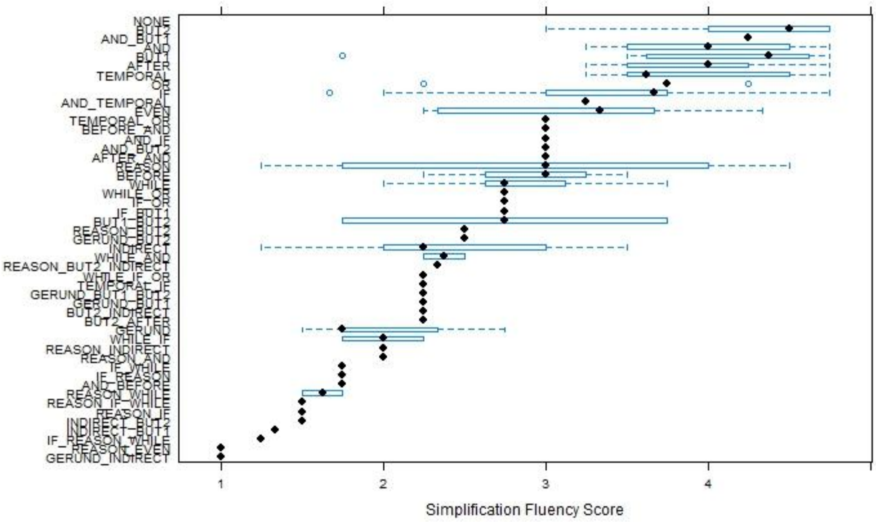


Figure 6.2: Simplification Fluency by evaluation dimension box-plots

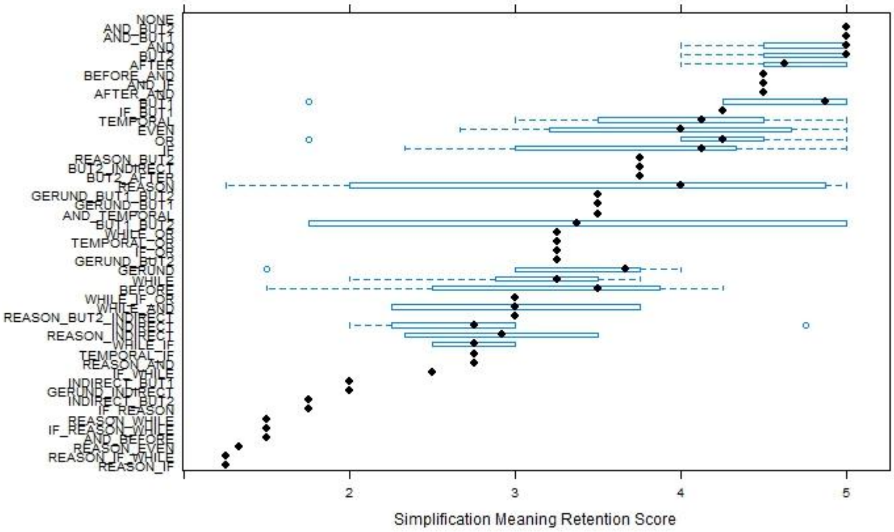


Figure 6.3: Simplification Meaning Retention by evaluation dimension box-plots

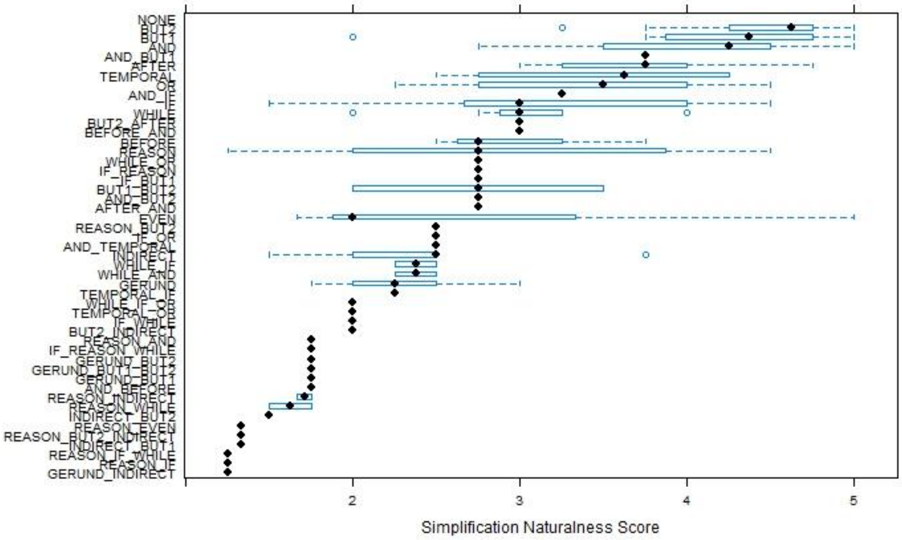


Figure 6.4: Simplification Naturalness by evaluation dimension box-plots

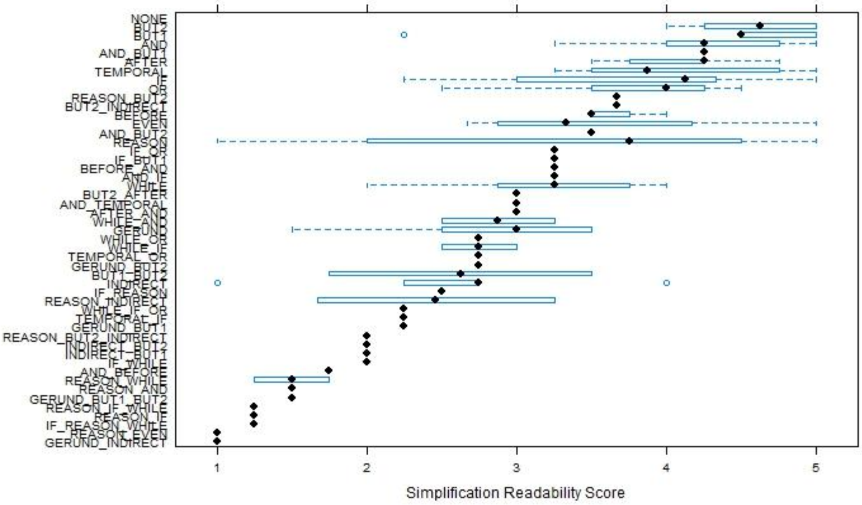


Figure 6.5: Simplification Readability by evaluation dimension box-plots

As the ANOVA results and the above graphs show, clearly not all clause reduction operations are created equal in terms of generating high-quality Korean sentences. Specifically, we see that, while some factorial combinations are consistently evaluated rather high, such as the BUT1_BUT2 grouping, the majority of combinations receive rather low scores. Factorial simplification seems like a blunt instrument, helping us as much as hurting us, and for future research we leave the development of a more controlled and precise tool. It is encouraging to note that, while there are a few exceptions, in general we see sentences bearing one simplification operation are rated quite high, particularly in terms of meaning retention. The usefulness of factorial sentence complexity reduction on MT will be discussed in further detail in the next two sections, 6.3 and 6.4.

6.3 Simplification + Machine Translation Results

Once translation evaluations were returned and the means calculated, overall translation quality increase, relative to the original, was again examined. In the case of Naver Translate, our system improved output 63% of the time. For Google Translate, simplification improved output 62% of the time. And finally, for Moses, improvements were seen 72% of time. This is summarized in Figure 6.6 below.

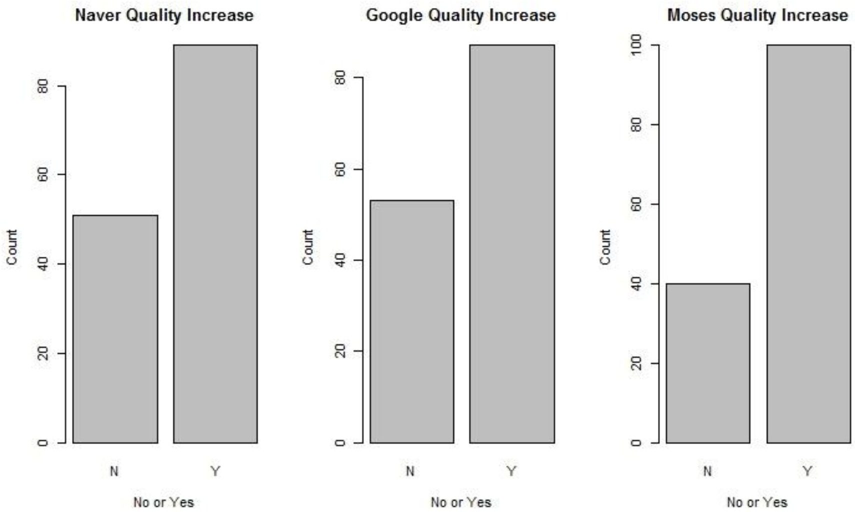


Figure 6.6: Naver, Google, Moses overall translation quality increase

This is an interesting result considering, at the time of writing, Google is an SMT system while Naver Translate employs state of the art neural networks, yet baseline Google Translate, without the help of simplification, slightly outperforms Naver Translate. It is, however, no surprise that both translation systems outperform our baseline Moses build.

For a more detailed look into which complexity reduction operations improve overall translation quality, please consider the following graphs.

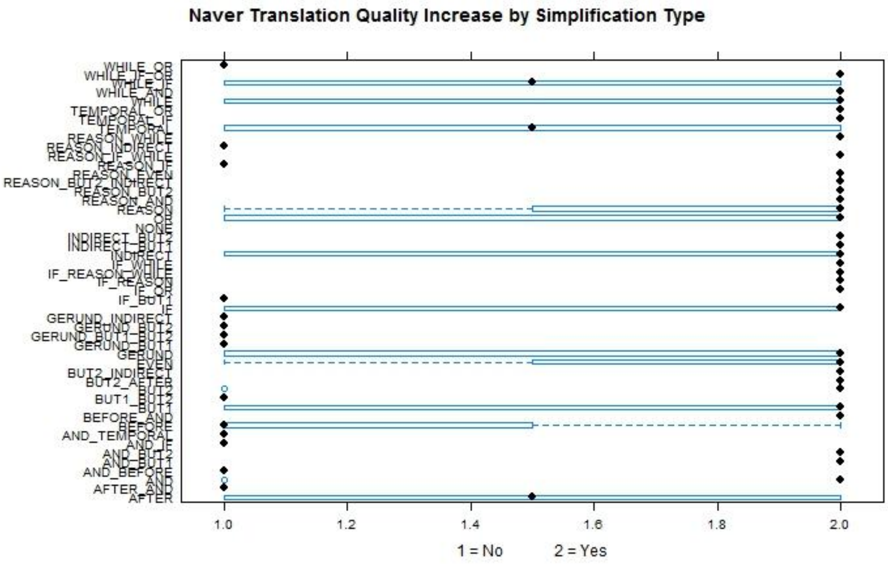


Figure 6.7: Naver Translate quality increase by complexity reduction operation

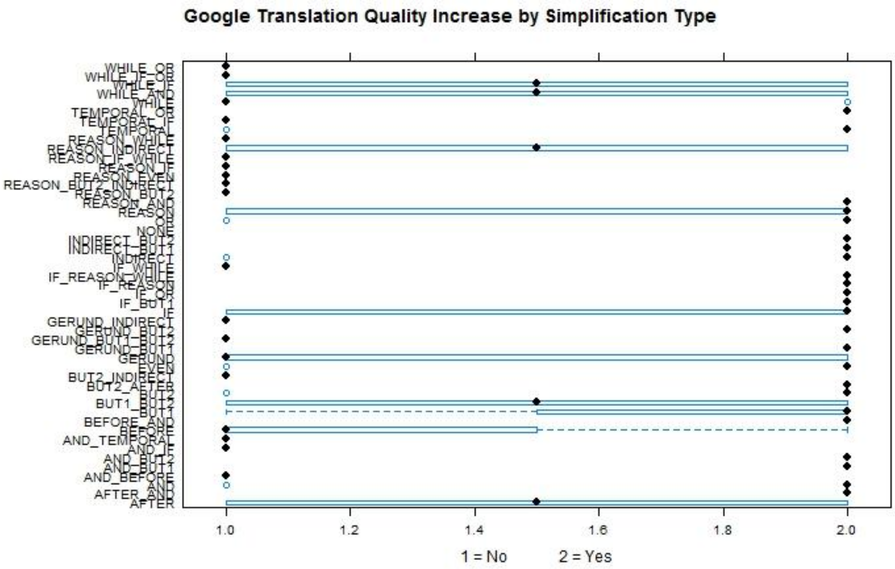


Figure 6.8: Google Translate quality increase by complexity reduction operation

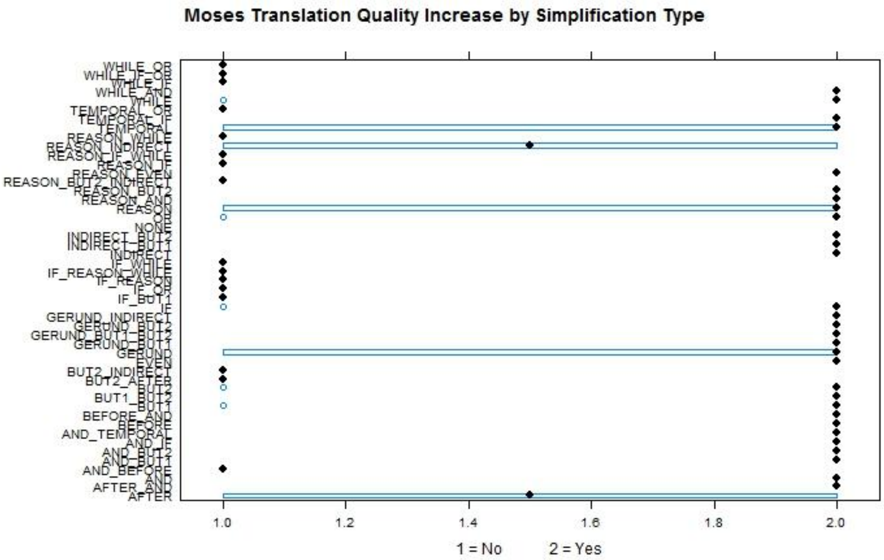


Figure 6.9: Moses translation quality increase by complexity reduction operation

In the above graphs, we see a recurring theme of single simplifications consistently resulting in translation improvements, excluding the *after* grouping, which consistently improves and does not improve translation quality between all three MT systems. While there are a few exceptions, such as the WHILE_OR and AND_BEFORE simplifications, there are very few sentences bearing multiple simplifications that consistently hinder translation quality between all three systems. For example, the sentence bearing a REASON_IF_WHILE complexity reduction operation did not improve output for the SMT systems Google Translate and Moses, yet it did for Naver Translate, an NMT system. Notice also that this is a combination of three simplification operations performed on one sentence, a complex operation that we would expect to decrease translation quality for an SMT system, but instead seems to benefit an NMT system. This will be discussed in greater detail in section 6.4 and 6.5.

Following this, ANOVAs were calculated to determine if the four evaluation metrics and translation quality increase were somehow related, and if so, was one dimension more related than the others? ANOVAs yielded significant results for Naver Meaning Retention and all other evaluation metrics. Results are provided in Table 6.3 below. For space considerations, in the following graphs only meaning retention will be considered, as this is the only metric that consistently yielded significant results between all three MT systems.

Dimension	**P-Value**
Naver Fluency	0.744
Naver Meaning Retention	**.0441**
Naver Naturalness	.692
Naver Readability	.107
Google Fluency	**.0128**
Google Meaning Retention	**1.01e-06**
Google Naturalness	**.00929**
Google Readability	**.000766**
Moses Fluency	**.00323**
Moses Meaning Retention	**1.16e-05**
Moses Naturalness	**.00628**
Moses Readability	**1.62e-05**

Table 6.3: Translation quality increase by evaluation dimension

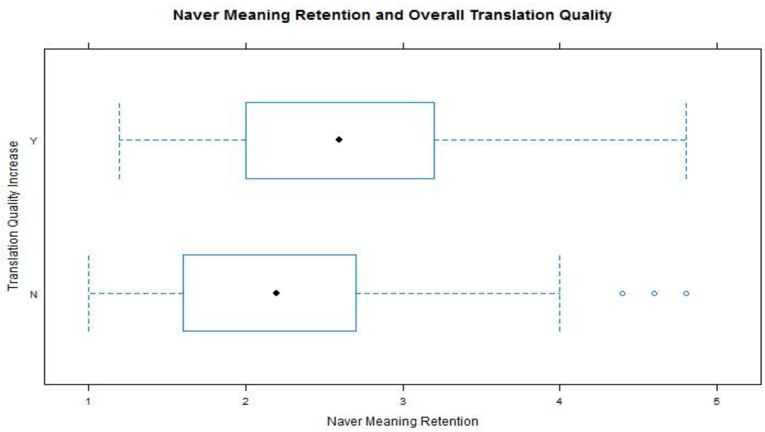


Figure 6.10: Translation quality increase by Naver Meaning Retention box-plot

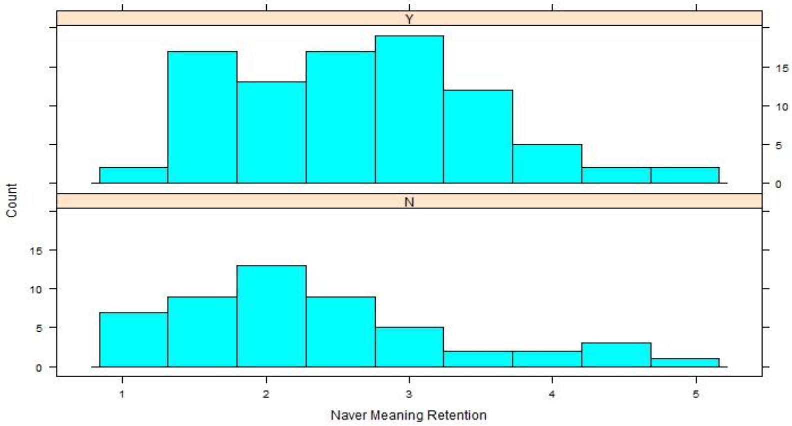


Figure 6.11: Translation quality increase by Naver Meaning Retention histogram

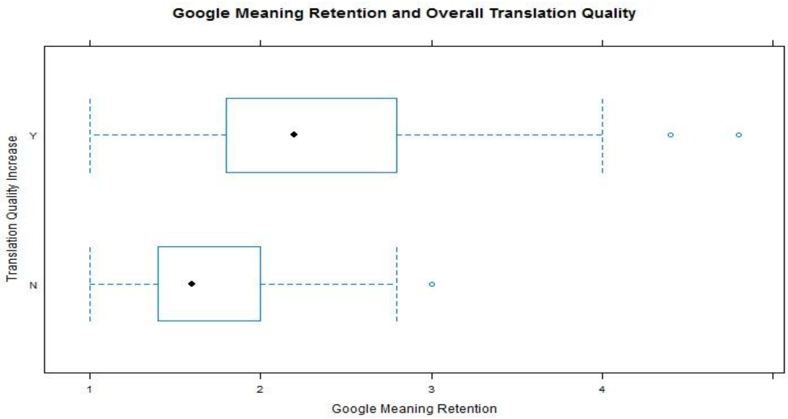


Figure 6.12: Translation quality increase by Google Meaning Retention box-plot

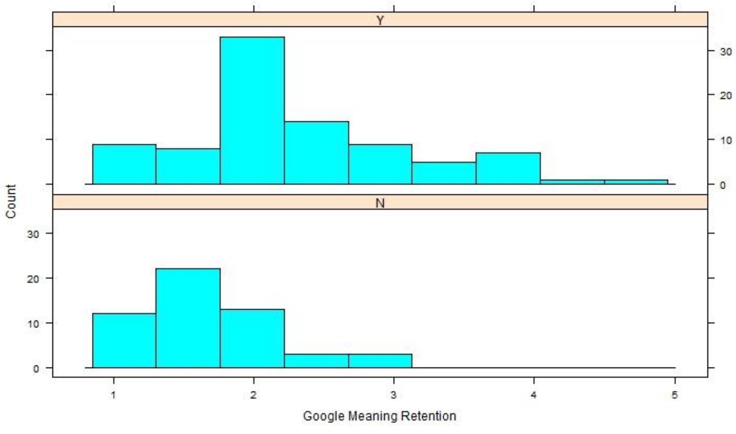


Figure 6.13: Translation quality increase by Google Meaning Retention histogram

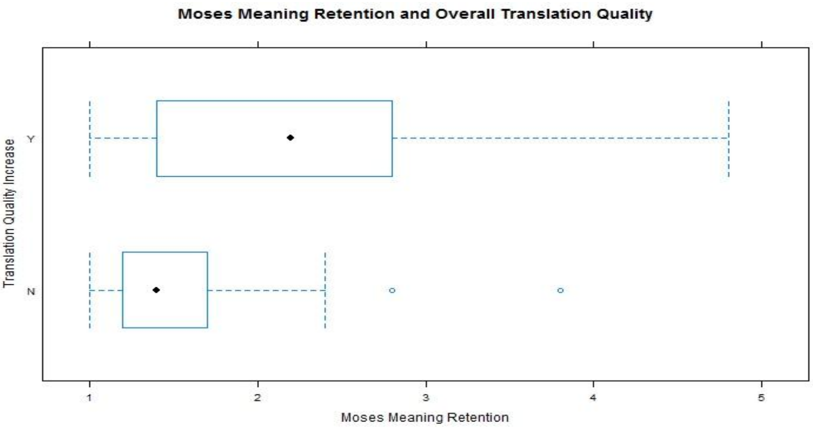


Figure 6.14: Translation quality increase by Moses Meaning Retention box-plot

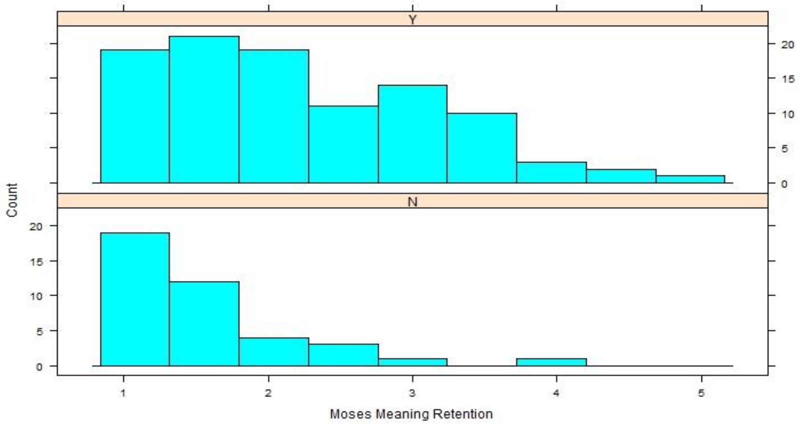


Figure 6.15: Translation quality increase by Moses Meaning Retention histogram

As the above ANOVA results and graphs show, meaning retention and translation quality increase seem to significantly influence one another. That is, as a given translation's meaning retention score increases, we would then expect to see more evaluators rate it as an improved translation.

If meaning retention score is predictive of translation quality increase, then what purpose do the other evaluation metrics serve? Recall that this is the first known TS study to use the binary metric of yes-no translation improvement, while other studies, such as Collados, used only the four evaluation dimensions standard to TS (2013). However, as can been seen in Table 6.3 above, if the three other evaluation dimensions are significantly different due to translation improvement, then we can expect meaning retention to mirror this difference. However, the reverse is not true, as can be seen with Naver Meaning Retention. If meaning retention is the only evaluation metric predictive of translation quality improvement, then the three other dimensions serve no purpose.

This research therefore suggests that, if the purpose of a given TS system is the improvement of MT system output, only the use of the binary yes-or-no metric, introduced in this study, and meaning retention to measure translation quality increase is necessary. This will reduce the burden on evaluators and make for a more streamlined and simpler process during evaluation and analysis.

Finally, ANOVAs were performed to determine if the proposed Korean sentence complexity reduction system as a preprocessing task for MT had a significant impact on the four evaluation dimensions. The same MT vs. SIMP + MT binary grouping from Chapter 5 was created. The results of the ANOVAs can be found in Table 6.4 below. The Naver Translate results will not be discussed any further in this section due to a complication that will be discussed at length in section 6.4. With that in mind, only Moses yielded statistically significant results, and keeping with the theme of meaning retention and limited space, only graphs reflecting Moses meaning retention will be shown, however, please note that the other Moses evaluation dimensions yield similar results.

Dimension	**P-Value**
Naver Fluency	**.000247**
Naver Meaning Retention	.231

Naver Naturalness	**.000168**
Naver Readability	**.000269**
Google Fluency	.58
Google Meaning Retention	.534
Google Naturalness	.846
Google Readability	.475
Moses Fluency	**.0471**
Moses Meaning Retention	**.0244**
Moses Naturalness	**.042**
Moses Readability	**.0102**

Table 6.4: Evaluation dimension compared against MT vs. SIMP + MT ANOVA results

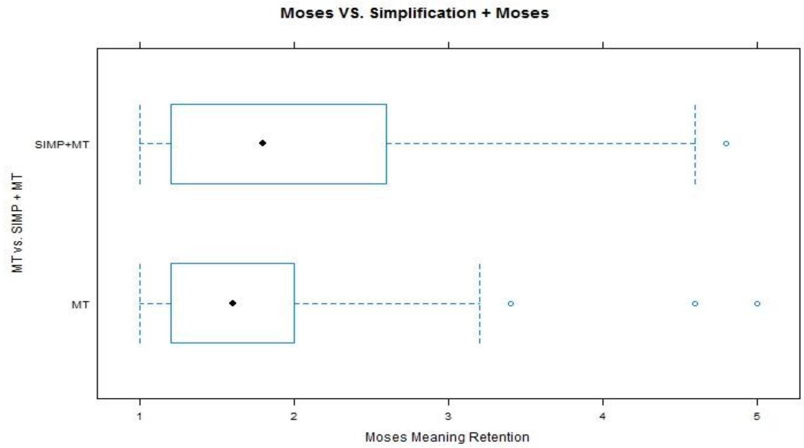


Figure 6.16: Moses meaning retention against MT vs. SIMP + MT box-plot

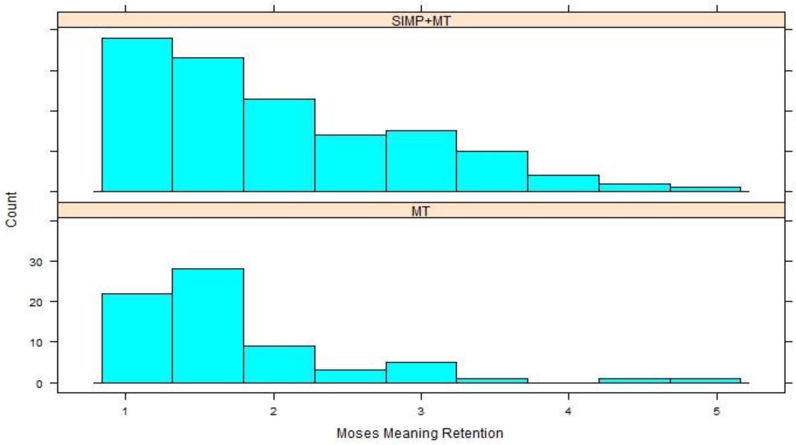


Figure 6.17: Moses meaning retention against MT vs. SIMP + MT histogram

As the above results show, Korean simplification as a pre-processing task for MT has a positive effect on MT system output. Though both MT and SIMP + MT show a great deal of left skewing in Figures 6.16 and 6.17, SIMP + MT has a much greater variance and achieves high evaluation scores more consistently than pure MT. Unfortunately, this result was not reflected in Google Translate, but this could be due to a number of reasons. For example, over 700 translations were evaluated by volunteer participants in this experiment, a scale never before attempted in the TS literature. While we had hoped this would give variety to our evaluations, it is very possible this instead resulted in evaluator fatigue. This will be discussed in further detail in section 6.5.

6.4 Naver Translate

It was not originally the intention of this study to use Naver Translate, however, at time of writing the NMT system only recently transitioned to neural network motivated machine translation. As such, it seemed worthwhile to use the new technology to comment on TS as a preprocessing task for NMT systems. One issue arose, however, during the evaluation of Naver Translate output. Recall from section 6.3 that an ANOVA comparing meaning retention scores between pure Naver NMT and SIMP + NMT yielded a p-value of .231, which is visualized below in Figure 6.18 and 6.19 below:

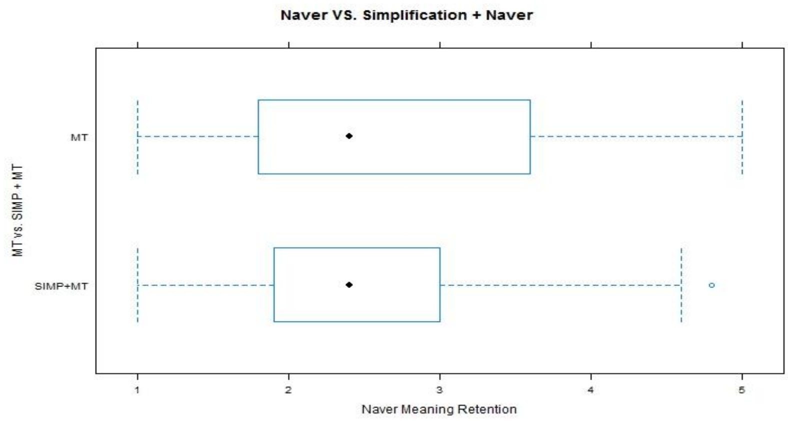


Figure 6.18: Naver meaning retention against MT vs. SIMP + MT box-plot

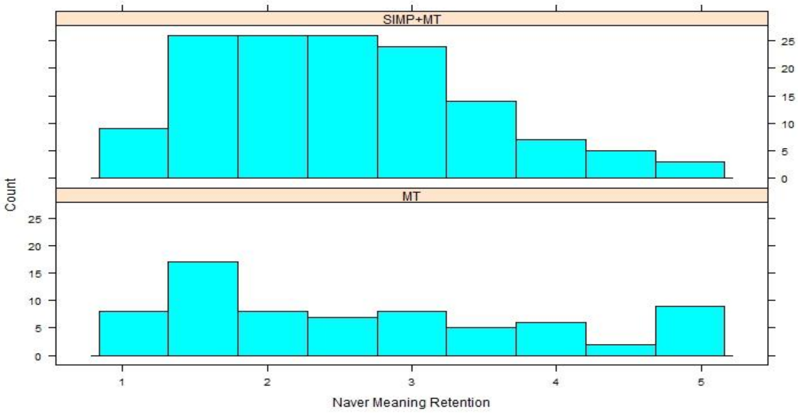


Figure 6.19: Naver meaning retention against MT vs. SIMP + MT histogram

If these graphs and results are to be believed, then it seems the proposed Korean sentence complexity reduction system actually has a negative effect on NMT systems. However, it was discovered that of the 70 original sentences extracted from the corpus, 10 were translated perfectly, relative to the desired translation, by the NMT system. Some the desired translations were of questionable English quality, so it seemed more likely that, instead of translating these 10 sentences perfectly (too perfectly) unaided on the first try, the Naver NMT system was likely trained on some of the sentences contained within the Samsung Corpus

(SC).

As was mentioned in Chapter 3, the exact original of the SC is not well understood, and since bilingual Korean-English data is not overly common, it does not seem unreasonable to assume that there may have been some overlap in data when Naver Translate was trained.

Believing that the data had been skewed, these 10 sentences, their simplifications, and their subsequent translations were therefore removed from the data set, resulting in a total of 174 translations from Naver Translate. Reanalysis on Naver Translate was performed. First we notice that evaluators indicated that sentence complexity reduction as pre-processing task for Naver Translate improves output 76% of the time, visualized below in Figure 6.20:

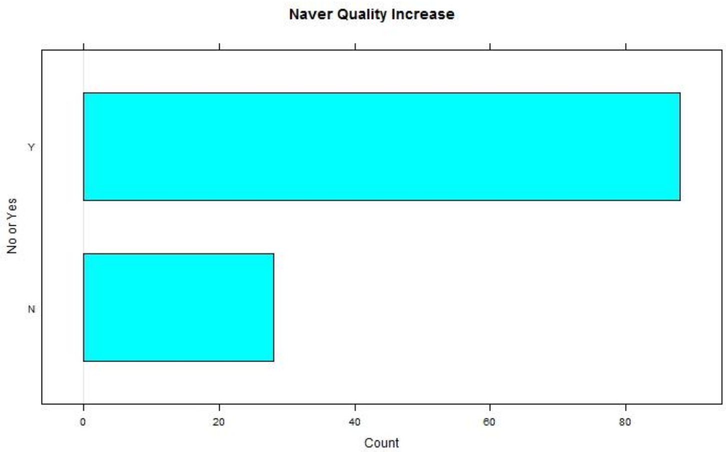


Figure 6.20: Naver Translate Quality Increase after simplification

The ANOVAs comparing evaluation dimension scores between pure Naver NMT and SIMP + NMT were performed again as well. All ANOVAs, except Naver Translate Fluency, yielded a significant result at p = 0.1, which can been seen below in Table 6.5. Keeping with the theme of meaning retention, Figures 6.21 and

6.22 will only reflect this evaluation dimension.

Dimension	P-value
Naver Fluency	.325
Naver Meaning Retention	**.0801**
Naver Naturalness	**.0539**
Naver Readability	**.0918**

Table 6.5: Naver Translate reanalysis: ANOVA results.

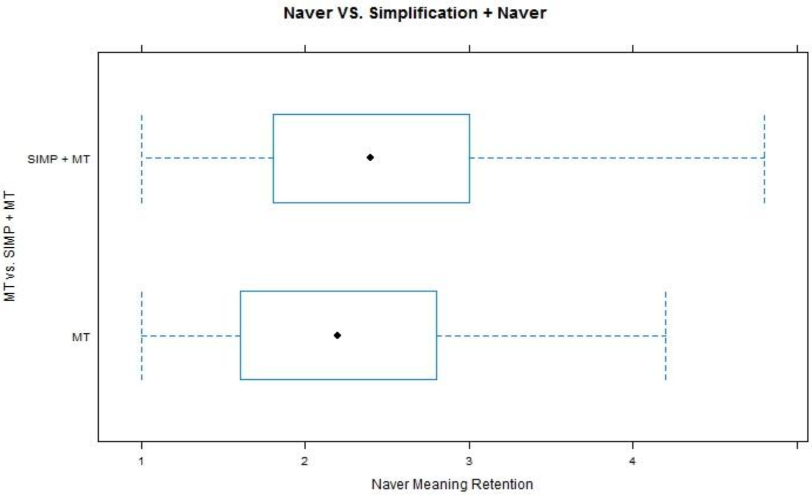


Figure 6.21: Naver Translate reanalysis: meaning retention against MT vs. SIMP box-plot

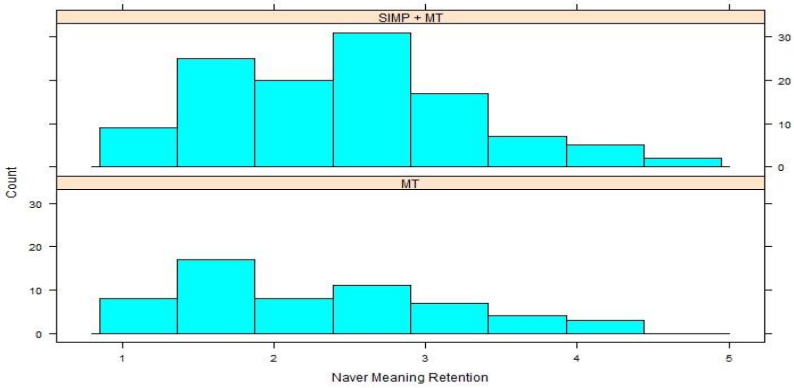


Figure 6.22: Naver Translate reanalysis: meaning retention against MT vs. SIMP histogram

Though not an overwhelmingly astounding result, as can be clearly seen from the above graphs, the proposed Korean sentence complexity reduction system as a preprocessing task, even for cutting-edge NMT systems, seems to improve translation quality. The variance of the distribution is much tighter and right-shifted in both Figures 6.21 and 6.22, indicated a more predictable and highly rated result when the output in question has been preprocessed by the proposed system. Neural networks are often touted as the harbingers of the age of artificial intelligence, so it is surprising to see such a powerful NMT system benefit from a rule-based simplifier. However, impressive as neural networks may be, they still have their foundation in statistical machine learning, which is based on frequency of occurrence and complex calculations. Both of these constraints are made easier and satisfied by a TS pre-processing step, as the current research proposes. This, in turn, results in improved NMT output.

6.5 Discussion

As was demonstrated and discussed in the preceding sections, the proposed Korean sentence complexity reduction system is capable of outputting relatively high-quality simplified sentences. In particular, the system shines when only a single operation is performed on sentences. When multiple operations are performed, output dramatically begins to shift into sounding unnatural. If the goal in mind is the production of high-quality simplified Korean sentences, the factorial-combination approach implemented in the current research may be too blunt a tool.

In terms of improving MT output, it has so far been demonstrated that Korean MT systems do indeed benefit from the proposed system. This research consistently demonstrated that Moses translation quality, the current academic standard, can be consistently improved by the proposed system. While the results were not quite as consistent for Google Translate, we believe this may have more to do with volunteer evaluators and the overwhelming number of evaluations we demanded of them.

MT quality improvement through simplification is not limited only to SMT systems, such a Google Translate and Moses, but the Naver Translate NMT system has also been demonstrated to benefit as well. This is consistent with what we would expect from the TS literature. In addition, this is the first known TS study to use TS in combination with an NMT system.

Though this research examined only Korean to English MT, it is our belief that other target languages would also benefit from Korean simplification. It was also demonstrated that, when the goal is the improvement of MT system output, the use of four evaluation dimensions is completely unnecessary and creates too great a burden on evaluators. Instead the use of meaning retention and binary yes-or-no translation quality improvement metric is more than sufficient for evaluating MT quality.

The factorial approach implemented in this study was very helpful for determining which individual simplification operations in complex sentences resulted in natural Korean and improved MT output, however, when multiple factorial permutations were performed on a single sentence, translation quality began to fade for SMT systems. However, consider table 6.6 below.

Number of Simplifications Performed	Naver Translate Quality Improvement?		Google Translate Quality Improvement?		Moses Quality Improvement?	
	Y	N	Y	N	Y	N
1	65	35	68	32	79	21
2	20	15	18	17	20	15
3	4	1	1	4	1	4

Table 6.6: Translation quality improvement by number of complexity reduction operations

We see that in terms of pure, raw counts, one simplification consistently results in translation improvement for all three MT systems. Two simplifications also consistently increase translation quality, but the counts between yes and no are no longer so vast and the difference is much closer. For the SMT systems Google Translate and Moses, three simplifications begin to become a hindrance and MT quality is no longer improved. However, we see the opposite for the Naver Translate NMT system, where more simplifications seems to increase the quality of output. In this way it seems that NMT systems behave in a fundamentally different way than SMT systems, so their behavior with TS is an interesting question we leave to future research. As was stated earlier, the number of simplifications was not controlled for in this study, so we hesitate to comment on it beyond our brief observation and exploration here and in section 5.4.

7. Conclusion

Text simplification as natural language processing task has a long history of improving machine translation output using European languages as the source language, but this is the first study to suggest the use of text simplification techniques with Korean as the object language. As such, we focused our efforts on the most rudimentary simplification technique in syntactic simplification: sentence complexity reduction through the splitting and rephrasing of syntactically complex Korean sentences bearing more than one clause.

This was accomplished through the construction of a supervised, rule-based sentence complexity reduction system for Korean. Of course, rule formation varied with each structure, however, it was our intention to generate natural Korean output that would also increase the translation quality of a machine translation system relative to the original, complex sentence. As this is the first study to implement a natural language processing motivated Korean text simplification system, only the first steps were taken and tested, and we leave the improvement, adaption, and expansion of this system for future research.

Though heavily influenced by the existing rule-based systems, unique steps were taken during the implementation of the proposed system. For example, within this system we made use of a phrase-grouping and generalization technique for nuance structures. By generalizing to a less productive, more common word-form, which bears relatively the same meaning as the original word-form, as a tool for disambiguation, we believe machine translation improvement can be achieved. Not only that, but this seems like a logical approach for most natural language processing tasks to cope with data scarcity. Additionally, hoping to overcome the initial hurdle of syntactically simplifying sentences bearing more than two clauses, our factorial complexity reduction system provides a unique and simple solution for the combination of multiple simplification operations within a sentence.

As this approach generates all possible simplifications and all possible simplification combinations, we hoped to shed light on which simplification types and which permutations yielded the most natural Korean and/or improved machine translation output. In order to asses system output, native speakers were recruited to numerically evaluate the quality of simplified Korean sentences. This is the first study to run a parallel comparison between simplified and translated text, as the majority of known text simplification research is researcher-intuition motivated.

Though helpful part of time, the factorial approach may be too blunt a tool for generating fluent Korean output that has been simplified multiple times. However, as this was the first study to attempt such an approach, particularly on Korean, we believe this contribution is meaningful. Additionally, as there is no previous Korean research to compare this work to, it is difficult to say whether the factorial approach is inappropriate or not. We leave for future research the implementation of a more precise tool for the generation of more fluent Korean output and comparison with other approaches.

If the goal is machine translation, fluent simplified output is not necessary, so long as translation quality is improved. Through the use of three machine translation systems, namely Naver Translate, Google Translate, and Moses, and two different experiments, this research has demonstrated that our system does indeed improve translation quality compared to an unaided system. To the best of our knowledge, this is the first text simplification study to examine the output of three different translation systems simultaneously, while also being the first study to examine the interaction of text simplification and neural machine translation. On top of this, we conducted a parallel human evaluation study on a scale never before attempted in text simplification. We had hoped the scale would provide us with consistent evaluations, however, due to the volunteer nature and the scale of the experiment, we suspect it may have resulted in evaluator fatigue. We leave for future research human evaluation in a more controlled environment.

Additionally, we suggest the abolishment of the four evaluation dimensions standard to text simplification when evaluating translated output, and instead suggest a binary yes-or-no translation quality improvement metric in combination with a numerical meaning retention dimension. This was demonstrated to be more than sufficient for assessing translation improvement, and would streamline evaluation and analysis, while also reducing the significant burden placed on evaluators.

In this research, we have demonstrated that the proposed system certainly has a positive effect on machine translation after one simplification, often two, however when three simplifications are performed, translation quality, like Korean quality, begins to break down in a statistical machine translation environment. However, interestingly, the same was not observed for Naver Translate, a neural-network motivated machine translation system, where evaluators indicated improvement in translation quality the vast majority of the time, even when three simplifications were performed. Yet, as the number of simplifications was not controlled for in this study, we will not comment on it further, leaving such research for the future. Clearly the interaction between text simplification and neural networks demands more research.

References

Ahmen, A. (2016). Google Translate now includes Sindhi and Pashto. Retrieved November 12, 2016, from http://www.brecorder.com/arts-a-leisure/lifestyle/279458-google-translate-now-includes-sindhi-and-pashto.html

Angrosh, M., Nomoto, T., & Siddharthan, A. (2014). Lexico-syntactic text simplification and compression with typed dependencies. In *COLING* (pp. 1996-2006).

Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*.

Candido Jr, A., Maziero, E., Gasperin, C., Pardo, T. A., Specia, L., & Aluisio, S. M. (2009, June). Supporting the adaptation of texts for poor literacy readers: a text simplification editor for Brazilian Portuguese. In *Proceedings of the Fourth Workshop on Innovative Use of NLP for Building Educational Applications* (pp. 34-42). Association for Computational Linguistics.

Canning, Y., Tait, J., Archibald, J., & Crawley, R. (2000). Replacing anaphora for readers with acquired dyslexia. In *Proceedings of the 4th Discourse Anaphora and Anaphor Resolution Colloquium (DAARC'00), Lancaster, UK*.

Chandrasekar, R., Doran, C., & Srinivas, B. (1996, August). Motivations and methods for text simplification. In *Proceedings of the 16th conference on Computational linguistics-Volume 2* (pp. 1041- 1044). Association for Computational Linguistics.

Chandrasekar, R., & Srinivas, B. (1997). Automatic induction of rules for t ext simplification. *Knowledge-Based Systems*, 10, 183–190. DOI: 10.1016/S0950-7051(97)00029-4

Cho, K., Van Merriënboer, B., Bahdanau, D., & Bengio, Y. (2014). On the properties of neural machine translation: Encoder-decoder approaches. *arXiv preprint arXiv:1409.1259*.

Chung, J. W., Min, H. J., Kim, J., & Park, J. C. (2013, June). Enhancing readability of web documents by text augmentation for deaf people. In *Proceedings of the 3rd International Conference on Web Intelligence, Mining and Semantics* (p. 30). ACM.

Clark, H. H., & Clark, E. V. (1968). Semantic distinctions and memory for complex sentences. *The Quarterly Journal of Experimental Psychology*, *20*(2), 129-138.

Coldewey, D. (2016). Google unleashes deep learning tech on language with Neural Machine Translation. Retrieved November 12, 2016, from https://techcrunch.com/2016/09/27/google-unleashes-deep-learning-tech-on-language-with-neural-machine-translation/

Collados, J. C. (2013). *Syntactic Simplification for Machine Translation* (Master's dissertation).

Google Translate. (n.d.). Retrieved November 12, 2016, from https://translate.google.ca/m/translate.

Han, C. H., & Han, N. R. (2001). Part of speech tagging guidelines for penn korean treebank. *IRCS Technical Reports Series*, 24.

Haspelmath, M., & Sims, A. (2013). *Understanding morphology*. Routledge.

Hutchins, W. J., & Somers, H. L. (1992). *An introduction to machine* *translation* (Vol. 362). London: Academic Press.

Inui, K., Fujita, A., Takahashi, T., Iida, R., & Iwakura, T. (2003, July). Text simplification for reading assistance: a project note. In *Proceedings of the second international workshop on Paraphrasing-Volume 16* (pp. 9-16). Association for Computational Linguistics.

Jurafsky, D., & Martin, J. H. (2014). *Speech and language processing*. Pearson.

Kamran, A. (2013). Hybrid Machine Translation.

Koehn, P., Hoang, H., Birch, A., Callison-Burch, C., Federico, M., Bertoldi, N., & Dyer, C. (2007, June). Moses: Open source toolkit for statistical machine translation. In *Proceedings of the 45th annual meeting of the ACL on interactive poster and demonstration sessions* (pp. 177-180). Association for Computational Linguistics.

Koehn, P. (2009). *Statistical machine translation*. Cambridge University Press.

Lee, H. G., Lee, J., Kim, J. S., & Lee, C. K. (2015, October). Naver machine translation system for wat 2015. In *Proceedings of the 2nd Workshop on Asian Translation (WAT2015),(*pp. 69-73).

Li, S., Wong, D. F., & Chao, L. S. (2012, July). Korean-Chinese statistical translation model. In *2012 International Conference on Machine Learning and Cybernetics* (Vol. 2, pp. 767-772). IEEE.

Marcus, M. P., Marcinkiewicz, M. A., & Santorini, B. (1993). Building a large annotated corpus of English: The Penn Treebank. *Computational linguistics*, *19*(2), 313-330.

Na, Hwidong. (2015). Non-projective Parsing for Pre-ordering Statistical Machine Translation (PhD Dissertation).

Papineni, K., Roukos, S., Ward, T., & Zhu, W. J. (2002, July). BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting on association for computational linguistics* (pp. 311-318). Association for Computational Linguistics.

Rosa, R. (2013). Automatic post-editing of phrase based machine translation outputs. *Master's thesis, Institute of Formal and Applied Linguistics, Charles University, Prague*.

Schütze, H. (2008, July). Introduction to Information Retrieval. In *Proceedings of the international communication of association for computing machinery conference*.

Siddharthan, A. (2002, December). An architecture for a text simplification system. In *Language Engineering Conference, 2002. Proceedings* (pp. 64-71). IEEE.

Siddharthan, A. (2014). A survey of research on text simplification. *ITL-International Journal of Applied Linguistics*, *165*(2), 259-298.

Specia, L., Caseli, H.M., Pereira, T.F., Pardo, T.A.S., Gasperin, C., and Aluísio, S.M. (2009). Building a Brazilian Portuguese parallel corpus of original and simplified texts. In Alexander Gelbukh (ed), *Advances in Computational Linguistics,Research in Computer Science, 41, pp. 59-70.* 10th Conference on Intelligent Text Processing and Computational Linguistics (CICLing-2009). Mexico City.

Specia, L. (2010). *Translating from Complex to Simplified Sentences. 9th International Conference on Computational Processing of the Portuguese Language (Propor-2010). Lecture Notes in Artificial Intelligence, 6001, Springer, pp. 30-39*. Porto Alegre, Brazil.

Specia, L., Jauhar, S. K., & Mihalcea, R. (2012, June). Semeval-2012 task 1: English lexical simplification. In *Proceedings of the First Joint Conference on Lexical and Computational Semantics-Volume 1: Proceedings of the main conference and the shared task, and Volume 2: Proceedings of the Sixth International Workshop on Semantic Evaluation* (pp. 347-355). Association for Computational Linguistics.

The National Institute of the Korean Language (2008) *Hangeul*, accessed November 2013, http://www.korean.go.kr/eng_hangeul.

Wołk, K., & Marasek, K. (2015). Neural-based machine translation for medical text domain. Based on European Medicines Agency leaflet texts. *Procedia Computer Science*, *64*, 2-9.

Wubben, S., Van Den Bosch, A., & Krahmer, E. (2012, July). Sentence simplification by monolingual machine translation. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics: Long Papers-Volume 1* (pp. 1015-1024). Association for Computational Linguistics.

네이버 번역기 (NAVER Korean Translator). (n.d.). Retrieved November 12, 2016, from http://translate.naver.com/.

초록

기계번역을 위한 한국어 문장 복잡성 감소

자연어처리 연구에서 성능 향상을 위해 사용되어온 전처리 작업 중 하나인 텍스트 단순화는 유럽어를 대상으로 오랫동안 연구가 되어 왔지만 아직 한국어를 대상으로 진행된 연구는 없었다. 그러나 한국어에서 영어로의 기계 번역과 같이 두 언어의 서로 다른 특성으로 인해 좋은 결과를 얻지 못하는 경우에 텍스트 단순화 연구의 필요성이 높아지고 있다.

한국어에서 영어로의 번역 품질 개선을 위하여 본 연구에서는 규칙 기반 텍스트 단순화 문제를 최초로 연구했다. 이 연구에서는 "뉘앙스 문법 구조의 구문 그룹화 및 일반화"라고 불리는 독창적인 기술 이 문제를 해결하였다. 이 기술은 특정 언어에 구애 받지 않고 다양한 언어 자연어 처리에 적용할 수 있다는 장점이 있다. 뿐만 아니라, 어떤 텍스트 단순화와 어떤 문장 생성 규칙이 유창한 한국어를 생성하고 기계 번역 출력을 향상시키는가에 대한 토대를 마련하기 위하여 계승단순화 구문 생성에 대한 고유한 요인 접근법 또한 구현되었다.

본 연구에서 제안된 방법에 의한 번역의 품질 평가를 위하여 한국 원어민 화자을 대상으로 단순화된 한국어 텍스트에 대한 평가를 실시하였고, 동시에 영어 원어민 화자에게 번역을 평가하도록 하였다. 본 연구에서 실험을 위해 사용된 번역 시스템은 Google Translate,

Moses를 사용하여 자체적으로 구축한 번역 시스템 그리고 인공 신경 회로망 기계번역 체계인 네이버 번역이다. 본 연구는 텍스트 단순화가 인공 신경 회로망 번역 체계에서 어떤 영향을 주는지에 대해 그리고 기존 다른 번역시스템에서의 텍스트 단순화의 영향을 비교 분석한 최초의 연구라는 데 그 의의가 있다.

전반적으로, 본 연구에서 제안된 체계는 비교적 유창한 한국어를 생성해 내었지만, 텍스트 단순화를 수행하는 방법론적이고 recursive한 특성으로 인하여 한번 이상의 단순화 작업 후에는 문장의 품질이 악화되기 시작했다. 반면에 기계번역을 위한 전처리 작업으로 제안 된 이 연구 방법은 본 연구에서 사용된 세 가지의 기계번역체계 모두에 대해 번역 수행 품질을 향상 시켰다. 두 가지를 단순화를 시켜도 번역품질이 향상되었다.

본 연구에서 사용 된 통계적 기계번역 체계의 경우, 두번 이상의 단순화는 한국어 문장 품질뿐만 아니라 번역 품질 또한 악화시켰다. 그러나, 본 연구에서 사용된 인공 신경 회로망 기계번역 체계인 네이버 번역의 경우 통사론적으로 복잡한 문장에 세 가지의 단순화를 시키더라도 번역 품질 개선이 평가자들이 있었다. 본 연구는 인공신경 회로망 기반의 자연언어처리 연구들과 번역 시스템에 있어서 더 많은 텍스트 단순화 연구가 필요하다는 것을 보였다.

주제어: 텍스트 단순화, 기계번역, 한국어, 자연언어처리
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